Behavioral/Cognitive

The Mechanical Variables Underlying Object Localization along the Axis of the Whisker

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Rodents move their whiskers to locate objects in space. Here we used psychophysical methods to show that head-fixed mice can localize objects along the axis of a single whisker, the radial dimension, with one-millimeter precision. High-speed videography allowed us to estimate the forces and bending moments at the base of the whisker, which underlie radial distance measurement. Mice judged radial object location based on multiple touches. Both the number of touches (1–17) and the forces exerted by the pole on the whisker (up to 573 μN; typical peak amplitude, 100 μN) varied greatly across trials. We manipulated the bending moment and lateral force pressing the whisker against the sides of the follicle and the axial force pushing the whisker into the follicle by varying the compliance of the object during behavior. The behavioral responses suggest that mice use multiple variables (bending moment, axial force, lateral force) to extract radial object localization. Characterization of whisker mechanics revealed that whisker bending stiffness decreases gradually with distance from the face over five orders of magnitude. As a result, the relative amplitudes of different stress variables change dramatically with radial object distance. Our data suggest that mice use distance-dependent whisker mechanics to estimate radial object location using an algorithm that does not rely on precise control of whisking, is robust to variability in whisker forces, and is independent of object compliance and object movement. More generally, our data imply that mice can measure the amplitudes of forces in the sensory follicles for tactile sensation.

Introduction

Diverse mammals use whiskers for navigation (Vincent, 1912; Dehnhardt et al., 2001), texture discrimination (Carvell and Simons, 1990; von Heimendahl et al., 2007), and object recognition (Brecht et al., 1997; Anjum et al., 2006). Rodents move their array of 35 long mystacial vibrissae (whiskers) through space to locate objects (Hutson and Masterton, 1986; Knutsen et al., 2006; Mehta et al., 2007; O'Connor et al., 2010a). Whiskers are elastic, conical beams (Birdwell et al., 2007). Contact between whiskers and objects produces time-varying stresses at the base of the whiskers (Birdwell et al., 2007) that are transduced by mechanoreceptors in the follicles into action potentials (Zucker and Welker, 1969; Dörfl, 1982; Gibson and Welker, 1983; Szwed et al., 2003; Szwed et al., 2006; Stüttgen et al., 2008).

Whiskers move in the horizontal plane by pivoting around a point in the skin. Whisker-based sensation is therefore naturally represented in cylindrical coordinates (Knutsen and Ahissar, 2009). Rodents accurately discriminate object locations in the anterior–posterior dimension, even with a single whisker (Hutson and Masterton, 1986; Knutsen et al., 2006; Celikel and Sakmann, 2007; Mehta et al., 2007; O'Connor et al., 2010a). Rodents could solve this problem by extracting the azimuthal angle of the whisker base (θ_{base}) at which the whisker first touches the object (θ_{touch}). θ_{touch} could be computed from the timing of touch referenced against a whisker-position-related signal (von Holst and Mittelstaedt, 1950; Fee et al., 1997).

Object localization in the radial dimension (r) is less well understood. Radial distance could be derived from forces at a single follicle resulting from the interactions between whisker and object (Fig. 1*A*,*B*). Because the whisker is tapered (Ibrahim and Wright, 1975), the stresses in the follicle are strongly distance dependent (Solomon and Hartmann, 2006; Birdwell et al., 2007; Knutsen and Ahissar, 2009). We use the term "force-dependent radial distance measurement" to describe the situation in which estimates of force amplitudes underlie radial object localization. Alternatively, mice could avoid force measurements and derive radial distance based on θ_{touch} and triangulation (Solomon and Hartmann, 2011). In that case, radial object localization would require either contact between an object and multiple whiskers simultaneously (Fig. 1*C*) or single whiskers over multiple contacts accompanied by movements of the follicle (e.g., the body or the head; Fig. 1*D*). A binary representation of touch, without measurement of whisker force magnitude, would suffice for object localization. Distinguishing these mechanisms requires experimental situ-

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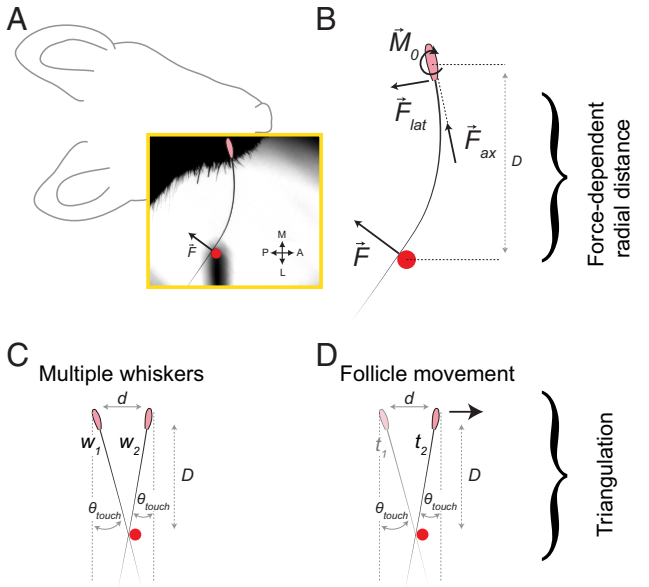

Figure 1. Radial object localization. Shown are the different strategies that could underlie radial object localization. ***A***, Schematic of the experimental situation treated in this study. A mouse protracts a single whisker (whisker C2) against a pole (highlighted in red) to determine its position. The pole exerts a force *F* on the whisker. ***B***, Mice could estimate radial distance from time-dependent or protraction-dependent stresses in the follicles (force-dependent radial distance measurement). These are produced by M_0, F_{ax}, and F_{lat}. ***C***, Triangulation with multiple whiskers. In this example, mice touch the object (red) with two whiskers (w_1, w_2), separated by *d*. The radial distance *D* can be extracted using θ_{touch} of both whiskers as $d/\tan(\theta_{touch}(w_1) - \theta_{touch}(w_2))$. ***D***, Triangulation with one whisker over multiple touches. In this example, mice touch the object two or more times (at times t_1 and t_2). The whisker follicle moves a distance *d* from time t_1 to t_2. The radial distance *D* can be extracted as $d/\tan(\theta_{touch}(t_1) - \theta_{touch}(t_2))$.

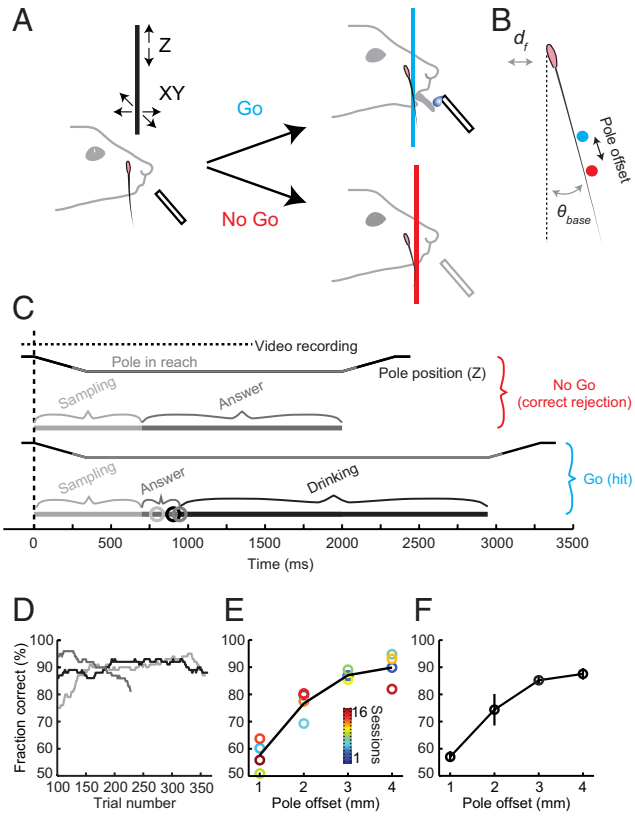

Figure 2. Behavioral analysis of radial object localization. ***A***, Schematic of the task. Left, The pole is out of reach. Right, The pole is within reach of the spared (C2) whisker. Mice judge the radial distance to the pole and respond according to the pole's location (Go stimulus, lick, blue; No Go stimulus, no-lick, red). ***B***, Top view of the pole positions. Trial types are defined by the location of the pole along the radial dimension (i.e., along the whisker). ***C***, Temporal structure of behavioral trials (top, correct No Go, "correct rejection"; bottom, correct Go, "hit"). During the sampling period (typically 0.7–1 s), mice explored the object with their whiskers and licking was not scored. During the answer period, licking was scored as a response (circles, answer licks from three typical trials; mean reaction time of three mice, five sessions, 635 ms). After the response, animals were allowed to drink (2 s). If the animal failed to respond, as in the No Go trial, the pole ascended 2 s after the beginning of the trial. ***D***, Performance during three typical behavioral sessions for three mice (fraction of trials correct, running average over 100 trials). ***E***, Radial distance discrimination as a function of offset between Go and No Go stimuli for one mouse (16 sessions). Circles indicate mean performance averaged over one session (colors indicate the sequence of behavioral sessions across time); black line, average. ***F***, Average performance (mean ± SD; $n = 5$ mice; 16–18 sessions per mouse). For some pole offsets, the error bars are smaller than the symbol.

ations in which θ_{touch} is nearly identical for different object locations.

We trained head-fixed mice to locate a pole with a single whisker (O'Connor et al., 2010a). The pole was placed at one of several locations along the whisker. High-speed whisker tracking (Knutsen et al., 2005; Voigts et al., 2008; O'Connor et al., 2010a) revealed the whisking strategies and forces supporting radial object localization. Our studies demonstrate that mice can perform force-dependent radial object localization and suggest that simultaneous measurement of lateral and axial stresses in the skin jointly underlie radial distance perception.

Materials and Methods

Mice. The behavioral procedures were similar to a previous study probing object localization in the anterior–posterior dimension (O'Connor et al., 2010a). Adult (> P60) male mice (C57BL/6Crl; Charles River Laboratories; $n = 8$) were housed individually in cages containing tunnels and bedding material in a reverse light cycle room. For 10 d before training and on days without behavioral testing, mice were maintained on 1 ml of water/d (O'Connor et al., 2010a). On days with behavioral sessions, mice obtained all water for the day during the session and performed until sated. The amount of water consumed was determined by weighing the mouse before and after the session. The volume consumed was ~1 ml. The weight and health of the mice were monitored daily. All procedures were in accordance with protocols approved by the Janelia Farm institutional animal care and use committee.

A custom head post was mounted to the skull under isoflurane anesthesia (~1.5–2% by volume in O_2). Bupivacaine or lidocaine HCl (10 μl, 0.5% s.c.) was applied at the incision site. The scalp and periosteum were removed from the dorsal surface of the skull. A thin layer of cyanoacrylate adhesive was applied to the skull and covered with dental acrylic (Lang Dental Jet Repair Acrylic 1223, catalog #1251546; Henry Schein). Head posts were fixed to the skull with dental acrylic. Buprenorphine HCl (0.05–0.10 mg/kg, i.p.; Bedford Laboratories) was used for postoperative analgesia. Ketoprofen (5 mg/kg) was administered at the time of surgery and postoperatively to control inflammation. Mice were allowed 10 d to recover from surgery before water restriction.

Apparatus. Mice judged the distance to a thin steel pole (diameter 0.4 mm; Fig. 2*A*). In some experiments (see Figs. 9, 10), a cleaned, straightened rat whisker was substituted as a "compliant" pole, which was deformed by contact with the mouse whisker. A pair of stepper motors (catalog #NA08B30; Zaber) and linear sliders (catalog #NDN 2–50.40; Schneeberger) moved the pole in the horizontal (*x*–*y*) plane. This assembly was mounted on a pneumatic linear slider (SLS-10–30-P-A Mini slide, catalog #170496; Festo), which rapidly (~250 ms) brought the pole into and out of reach of the whiskers along the *z*-direction, triggered by a computer-controlled solenoid valve (CPE 10-M1BH-5L-QS-6, catalog #196883; Festo). The entire behavioral apparatus was enclosed in a light- and sound-isolation box.

The apparatus was controlled by software (http://brodylab.princeton.edu/bcontrol). A real-time Linux system interfaced with valves and re-

corded licking responses using a PCI-6025E data acquisition board (National Instruments). The stepper motor was driven by its controller in response to commands sent through a serial port of a Windows computer running MATLAB.

A custom lickport, which provided the water reward and recorded licking, was placed within reach of the mouse's tongue. Licks were recorded as interruptions in the light path between an LED (860 nm, LN77L; Panasonic) and a phototransistor (L14G1; Fairchild Semiconductor). The phototransistor and LED signaled interruptions to the control computer.

Water was delivered by gravity into the lickport through a 1.3-mm-diameter steel tube under solenoid valve control (The Lee Company). Excess water was sucked out of the lickport through a tube (0.8 mm internal diameter [i.d.]) using a peristaltic pump (Rabbit Plus peristaltic pump; Rainin Instrument). Punishment was provided by puffs of compressed air (typically 10 psi) delivered through a small metal tube (~2.3 mm inner diameter) pointed at the face from a distance of several centimeters. The air was controlled by a solenoid valve (The Lee Company) connected to a compressed air source.

During behavior, mice were crouched in a natural position in acrylic (2.9 cm i.d.; catalog #8486K433; McMaster) tubes such that their heads protruded out at the front and they could use their front paws to grip the tube's edge. The headpost was secured to the bench using a kinematic mount, guaranteeing precise positioning of the head across sessions. Whiskers were free to move around the space surrounding the heads.

Imaging whiskers. A high-speed videography system using CMOS cameras (200 × 310 pixels, 500–1000 fps, 0.2 ms exposure, 8-bit depth, catalog #A504 K, Basler or Imaging Studio software and X-PRI camera, 180 × 250 pixels, 500 fps, 0.5 ms exposure, 8-bit depth, AOS Technologies) measured the position and shape of whiskers during behavior. Pixel size was either 0.07 or 0.06 mm. For every trial, a video of 1.5 s length was recorded starting 83 ms before the triggering of the pole descent (Fig. 2*C*). Illumination was with an infrared LED (940 nm, catalog #ELJ-940–211; Roithner Lasertechnik). The light was delivered through a diffuser and a condenser lens and pointed directly into the camera.

Whisker trimming and manipulation. Whiskers were gradually (over 12–27 d) trimmed down to the C2 whisker under light isoflurane anesthesia. Whiskers were retrimmed every 2–4 d. To stiffen whiskers, three cut whiskers from the contralateral side were glued to the principal whisker with hairspray (göt2be glued Freeze Spray; Schwarzkopf). Two of these whiskers had their tips facing toward the base of C2 and the third in the same orientation as C2, producing a thickened whisker without taper. After the experiment, the whiskers and hair spray were removed with soap and water. For control experiments (two sessions per mouse), the three whiskers were glued to the contralateral C2. The stiffening procedure has unintended side effects in addition to stiffening and abolishing taper: the hairspray might alter the friction coefficient of the whisker and also increases whisker weight.

Behavior. Head-fixed mice judged the distance to a metal pole that was presented at one of two positions (Fig. 2*A*,*B*). The sound of the pneumatic slider controlling the vertical pole position indicated trial start. The pole locations were arranged to lie along the whisker in the radial dimension. A proximal position was defined as the Go position; licking and breaking the beam of the lickport triggered a water reward (scored as a hit). One or several distal positions were defined as No Go positions; licking and breaking the beam triggered a time-out (2–10 s) and an airpuff as mild punishment (false alarm). Trials in which mice did not lick were neither rewarded nor punished in both the Go (miss) and No Go (correct rejection) trials.

Licking triggered reward or punishment only during the answer period, which began some time after the pole became available for sampling (sampling period; Fig. 2*C*). When there was no lick response (correct rejections and misses), the pole ascended and the trial was over. The sampling period was adjusted across and during sessions (typically between 0.5 and 1.25 s) to improve the mouse's performance. The answer period ended 2 s after the start of the pole descent. During a hit trial, the water valve opened, dispensing a drop of water (~8 μl) and terminating the answer period. A drinking period was granted to allow the mouse to drink before the end of the trial. The probability of each of the two trial types was typically 50%, but was sometimes adjusted during training to suppress false alarms. To prevent discouraging mice with a succession of No Go trials, the number of successive trials of one type was limited to three.

Probing the mouse's ability to perform radial distance discrimination required that the azimuthal angle was as similar as possible for all object locations. The Go position (target) and No Go position (distracter) were selected based on the whisker video. The target was always 5 mm from the follicle. For the standard task, the distracter was 8 mm from the follicle (Fig. 2*D*). For the experiments shown in Figure 2, *E* and *F*, the distracter positions were in the range of 6–9 mm from the follicle; only one distracter position was used per session. For the experiments shown in Figure 7*E*–*G*, azimuthal jitter was introduced for both Go and No Go trials. In the absence of jitter, we defined a vector from the follicle to the point of contact between the whisker and pole (length $r = 5$ mm, Go; $r = 8$ mm, No Go). We then defined an arc by rotating the vector about the follicle by +3° and −3° (Fig. 7*E*). Pole positions across trials were then chosen randomly from points on that arc. For the experiments shown in Figures 9 and 10*C*, distracter positions were randomly chosen across trials from the range 7–13 mm, which generated a large range of bending moments and axial forces in the No Go positions. This large range of forces relaxed the demands on the properties of the flexible object for creating appropriate illusion trials.

To define the reaction time for a session (Figs. 2*C*, 6, 7, 8, 9, 10*C*), we constructed a histogram of the times of the first lick across trials. The reaction time was the time between the pole coming into reach (earliest frame of contact in the session, typically 250 ms) and the mode of the histogram. For individual hit and false alarm trials, the reaction time was defined as the time from earliest frame of contact in the session to the animal's first lick in that trial. Because mice did not lick in correct rejection and miss trials, we used the mean reaction time of the hit trials as a proxy. For correct rejection and miss trials, we defined the time point of the animal's decision as the time of first contact with the pole plus the mean reaction time on hit trials (time between first contact and first lick).

To determine the amplitude of whisking, we first filtered θ_{base} between 6 and 60 Hz (Butterworth filter) and then performed a Hilbert transformation on the filtered whisking signal (Huber et al., 2012). The amplitude component of the result is the whisking amplitude over time. For every trial, we determined the peak amplitude.

Measurement of whisker stiffness. Mice performed all experiments with a single C2 whisker. After conclusion of the behavioral experiments, we analyzed the structure of three whiskers with light and electron microscopy (Fig. 5*A*,*B*). The whisker radius was measured as a function of distance from the follicle. The taper was remarkably linear ($n = 3$, $r = -0.988$) and the whisker can thus be modeled as a cone. The second moment of inertia is as follows (Birdwell et al., 2007):

$$I_p = \frac{\pi}{4}(a_p)^4 = \frac{\pi}{4}\left(a_{base}\left(1 - \frac{s_p}{length_{whisker}}\right)\right)^4, \qquad \text{(M1)}$$

where a_{base} is the radius at the base of the whisker, s_p is the length along the contour of the whisker from the base to point p, and $length_{whisker}$ is the total length of the whisker.

The whisker-bending stiffness was probed by pushing the whisker, which was mounted on a micromanipulator by its root, against a metal pole on a high-resolution balance (Mx5; Mettler Toledo; readability 1 μg, repeatability at full load, 0.9 μg) in 1 mm increments along the whisker. The measured weights ranged from <1 μg (0.1 mm displacement 1 mm from the tip) to 630 mg (0.5 mm displacement at 10 mm from the follicle); 87% of measurements were >20 μg, and therefore have expected errors ≤2.5%. Force on the whisker was calculated as follows:

$$F_{scale} = mass(kg) \cdot 9.8 m/s^2. \qquad \text{(M2)}$$

The whisker contour was extracted from photographs. For small displacements (<1 mm), force scaled linearly with deflection.

The bending stiffness (EI_p) was estimated by comparing measured deflections to a numerical model (Birdwell et al., 2007). Model whisker deflection was iteratively calculated from contact point back to whisker

base in 1 μm segments using the following relationship (Solomon and Hartmann, 2006):

$$d\kappa_p = \frac{|\vec{r}_p \times \vec{F}|}{EI_p}. \quad \text{(M3)}$$

For each set of deflections at a horizontal distance x from the base, numerical estimates of whisker shape were generated for a range of E. The model error was calculated as the ratio (A2/A1) of the area (A2), between the model deflection and actual whisker deflection from base to pole, and the area (A1) between the actual whisker deflection and the undeflected whisker from base to pole (Birdwell et al., 2007). The E that minimized the error at each x was tightly clustered around 5 GPa for whisker JF25395 (Fig. 5*C*), and ranged from 4 to 8.5 GPa for whisker JF25403. We used $E = 5$ GPa to calculate whisker forces.

Calculation of forces based on measurements from high-speed video. In the main text, we derive the equations that allow forces to be calculated based on the idealized geometry of whisker and contacted object. However, in practice, critical details need to be considered in extracting these geometrical quantities from image data. In this section, we describe at the "algorithmic" level how we calculated forces, starting with the representation of the whisker as a sequence of points representing its medial axis (i.e., starting after "whisker tracking").

Tracking whiskers in raw video images was performed as described previously (O'Connor et al., 2010a; Clack et al., 2012). Equations given below for angles and follicle coordinates reflect the specific videographic conditions of our experiments, in which the mouse face appeared at the top of the image and protraction involved rightward motion of the whisker within the image. After tracking, the whisker's medial axis is represented as a sequence of n points (x_i,y_i). To overcome discretization noise for computing derivatives, the shape was approximated as a parametric curve:

$$c(l) = (x(l), y(l)), \quad \text{(M4)}$$

where $l \in [0,1]$ and $x(l)$ and $y(l)$ are fifth-degree polynomials. The polynomials were computed by fitting x_i and y_i, respectively, as a function of l_i, where, $l_1 = 0$ and:

$$l_i = \frac{\sum_{k=2}^{i} \sqrt{(x_k - x_{k-1})^2 + (y_k - y_{k-1})^2}}{\sum_{k=2}^{n} \sqrt{(x_k - x_{k-1})^2 + (y_k - y_{k-1})^2}}. \quad \text{(M5)}$$

For subsequent evaluation of $x(l)$ and $y(l)$, l was evaluated at 100 evenly spaced points in the interval [0, 1].

Arc-length distance along the whisker (s; in mm) is derived from $c(l)$ at each time point (i.e., for each video frame) as follows:

$$s(l) = \int_0^l \sqrt{x'^2 + y'^2}\,dl. \quad \text{(M6)}$$

Tracking of the face-most edge of the whisker can be unreliable due to movement of the fur on the whisker pad. In some early experiments, a small segment of whisker was obscured by the lickport. For these reasons we added a "mask" (Figs. 3, 4). For all analyses presented here, the mask consisted of a horizontal line chosen so that neither fur nor lickport would obscure the beginning of the whisker. This allowed the whisker to be effectively "truncated" at the intersection of a particular location on the whisker and the mask.

The arc-length distance at the intersection of the whisker and the mask was subtracted from $s(l)$, so the arc-length origin becomes the intersection of the whisker and the mask. Therefore, considering the mask, $s(l)$ was redefined as follows:

$$s(l) = \int_0^l \sqrt{x'^2 + y'^2}\,dl - \int_0^{l_{\text{intersection}}} \sqrt{x'^2 + y'^2}\,dl, \quad \text{(M7)}$$

where $l_{\text{intersection}}$ is the value of l at which $c(l)$ intersects the mask.

Angle (θ; in radians) at each point l along the whisker, for each time point, is as follows:

$$\theta(l) = \tan^{-1}\left(\frac{x'(l)}{y'(l)}\right), \quad \text{(M8)}$$

where x', y' denote derivatives with respect to l.

θ gives the azimuthal angle in the horizontal plane. Protraction corresponds to increasing θ. $\theta = 0$ is perpendicular to the midline of the mouse.

Angle at the whisker base (θ_{base}; in radians) at each time point is as follows:

$$\theta_{\text{base}} = \theta(l_{\text{base}}), \quad \text{(M9)}$$

where $l_{\text{base}} = \arg\min |s(l)|$.

Angle at the point of contact between whisker and pole (θ_{contact}; in radians) at each time point is as follows:

$$\theta_{\text{contact}} = \theta(l_c), \quad \text{(M10)}$$

where l_c is the value of l at which the whisker is closest to the center of the contacted pole, calculated for each time point as follows:

$$l_c = \arg\min\left(\sqrt{(x(l) - x_0)^2 + (y(l) - y_0)^2}\right), \quad \text{(M11)}$$

where (x_0,y_0) is the center of the contacted pole object. The point of finding the location along the whisker nearest the center of the contacted pole was: (1) to define operationally the location of contact during periods of contact and (2) to help identify such periods.

To help identify periods of whisker-pole contact, we calculated for each time point a quantity (d_{pole}; in mm) to estimate the distance between whisker and pole as follows:

$$d_{\text{pole}} = \sqrt{(x(l_c) - x_0)^2 + (y(l_c) - y_0)^2} - barRadius, \quad \text{(M12)}$$

where *barRadius* is the radius in pixels of the pole presented to the mouse.

The coordinates of the contact point were defined as follows:

$$(x_c, y_c) = (x(l_c), y(l_c)). \quad \text{(M13)}$$

Note that for the data presented here, (x_c, y_c) was not extrapolated to accommodate any failure of whisker tracking near the pole (the shadow of the pole could cause termination of whisker tracing at points very near the contact point). However, we have found this extrapolation useful in analysis of similar videography datasets and, to illustrate our general methods for extracting forces and moments from videography, we illustrate this step in Figure 3*E*. When extrapolated, Equations M11 and M13 were generalized to define the whisker-object contact point as the point along the whisker or its extrapolation that was closest to the center of the bar.

To define curvature, we could in principle use the "primary" fitted whisker, $c(l)$ in Equation M4. However, there can be variations in the extent of the whisker actually tracked from frame to frame, mainly because: (1) the field of view does not necessarily capture the entire whisker and the whisker can therefore be more or less fully in the field of view, and (2) the shadow of the pole could terminate tracing of the whisker near the whisker-pole intersection, such that the whisker was not traced distal to the pole. Therefore, $c(l)$ could be fitted to different segments of the whisker in different frames. This could generate subtle changes in the shape of $c(l)$ even at a fixed point on the whisker and thus spurious changes in the apparent curvature (Fig. 4). To mitigate this problem in measuring curvature, we fitted a "secondary" parametric curve as follows:

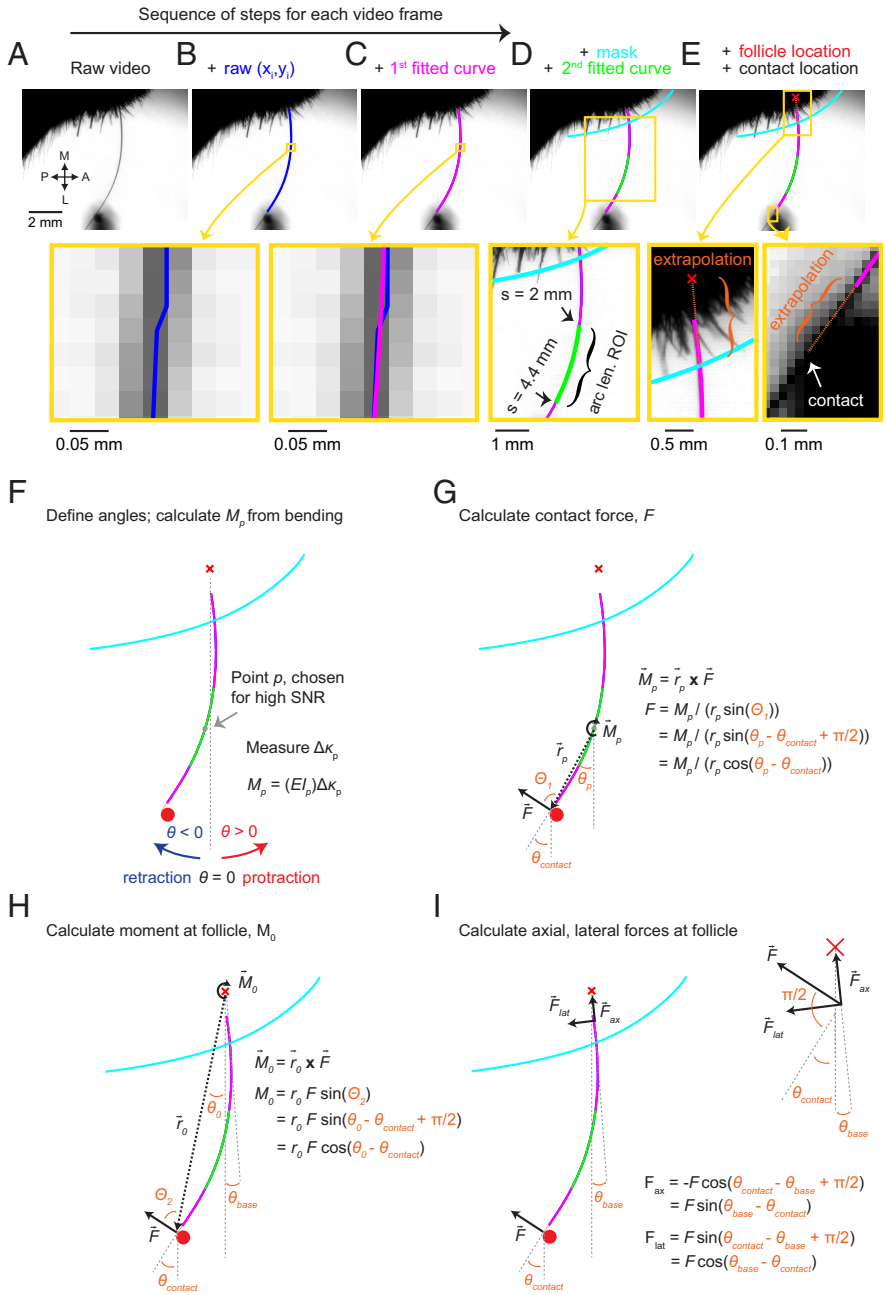

Figure 3. Derivation of bending moments and forces from single video frames. ***A***, Raw video frame (same as in Fig. 1*A*) showing a C2 whisker bent during contact with a pole (bottom view). The shadow of the mouse face is in the top left. All subsequent panels use the same video frame. ***B***, The tracker extracts a sub-pixel resolution series of points (x_i,y_i), representing the medial axis of the whisker. ***C***, The (x_i,y_i) points were fitted by a parametric curve (magenta) comprising fifth-order polynomials fitted to both *x*- and *y*-coordinates as a function of arc length along the whisker. Derivatives used in subsequent steps for computing angle at different points along the whisker were based on this fitted curve. ***D***, To avoid noise in whisker angle and length near the face induced by fur on the whisker pad, we effectively truncated the whisker at an intersection with a "mask' (cyan). Angle at the base of the whisker (θ_{base}) was estimated from the angle at the intersection between mask and whisker; this is warranted because the whisker is very stiff and straight close to the face. Similarly, the arc length origin was set to occur at this intersection, rather than at the first tracked point, which could vary depending on how far "into the fur" a whisker was tracked (see Fig. 4 for more details). A second-order polynomial was fitted to *x*- and *y*-coordinates in a small region of interest defined by arc length (***s***) along the whisker. This region of interest was chosen to maximize the signal-to-noise ratio in curvature estimates (green region in ***F***). Derivatives used in subsequent steps to measure whisker curvature were based on this "secondary" parametric curve. ***E***, The location of the whisker follicle (indicated by the red X) must be known to compute moment at the follicle. The follicle location was estimated by linearly extrapolating for a fixed distance past the mask based on the angle at the base of the whisker (left magnified box). The location of whisker-object contact was computed by linearly extrapolating beyond the last tracked point along the whisker and defining the contact point as the closest point along this line to the center of the pole (right magnified box). ***F–I***, Illustration of calculations. In these panels, the raw image from ***A–E*** is omitted, but all fitted quantities are shown: raw whisker (x_i,y_i) points (blue, but obscured because of successful fitting by the magenta curve), primary fitted curve (magenta), secondary fitted curve (green), mask (cyan), and follicle location (red X). The pole is illustrated schematically by the red circle at bottom. Images are slightly enlarged from those in (***A–E***) for clarity. ***F***, We defined azimuthal angles (θ) for the whisker and for vectors with respect to the axis perpendicular to the midline of the mouse (i.e., the medial-lateral axis; shown by gray dashed line). An angle of 0 implies parallelism with the medial-lateral axis. Protraction corresponded to increasing angle. The first step was to calculate the magnitude of moment at a point, *p*, along the whisker (M_p). This requires a measurement of change in curvature (from the undeflected case; $\Delta\kappa_p$) and the bending stiffness (EI_p) at *p*. In principle, we could have used any *p*, but we chose *p* for a high signal-to-noise ratio for curvature measurement. Because the whisker is linearly tapered, the larger curvatures occur farther from the whisker base, leading to more reliable measures of curvature. However, *p* should not be too close to the pole because the lever arm inducing curvature eventually vanishes and tracking errors near the intersection of whisker and pole can interfere with curvature measurement. Bending stiffness of individual whiskers was estimated using measurements on isolated whiskers (see Materials and Methods; Fig. 5). ***G***, The magnitude of the contact force (*F*), can be calculated from M_p, together with the length of the vector ($\vec{r}_p$) connecting *p* to the site of whisker-object contact, and the angle (Θ_1) between r_p and the contact force vector ($\vec{F}$). Because contact force is perpendicular to the whisker at contact (with negligible friction), Θ_1 can be obtained using only the angle of $\vec{r}_p$ (denoted θ_p) and the angle of the whisker at contact ($\Theta_{contact}$). ***H***, Having measured *F*, the moment at the follicle (M_0) can be obtained after measuring the length of the vector ($\vec{r}_0$) connecting the point of contact to the follicle, and the angle (Θ_2) between $\vec{r}_0$ and contact force vector $\vec{F}$. ***I***, Magnitudes of the axial (F_{ax}) and lateral (F_{lat}) components of force can be calculated as the length of the projection of $\vec{F}$ onto the long and short axes of the whisker at the follicle, respectively. Inset shows how the angles between vectors are computed using the angle of the whisker at base (θ_{base}) and the angle at contact ($\theta_{contact}$).

$$\hat{c}(l) = (\hat{x}(l), \hat{y}(l)) \qquad \text{(M14)}$$

to a segment of the whisker falling within a constant region of arc length (Fig. 4*D*); that is, where $\hat{x}(l)$ and $\hat{y}(l)$ are polynomials fitted to x_i and y_i, respectively, as a function of l_i, for *i* such that $a \leq s(l_i) \leq b$. Here we used second-degree polynomials fitted over the interval $[a,b]$. Therefore, $\hat{c}(l)$ was fitted over a constant length of whisker, although subject to imperfections due, for example, to pixilation and to variability in tracking the whisker near the face. The interval $[a,b]$ was chosen after inspection of the residuals between $\hat{c}(l)$ and the raw (x_i,y_i). Over small intervals, second-degree polynomial fits were simple and accurate.

At each time point, we derived curvature (κ_p; in mm^{-1}) from $\hat{c}(l)$ at a user-defined arc-length distance (in mm) along the whisker, s_p, as follows:

$$\kappa_p = \frac{\hat{x}'(l_p)\hat{y}''(l_p) - \hat{y}'(l_p)\hat{x}''(l_p)}{(\hat{x}'(l_p)^2 + \hat{y}'(l_p)^2)^{\frac{3}{2}}}, \qquad \text{(M15)}$$

for l_p such that $s(l_p) = s_p$.

Because whiskers can have intrinsic curvature, a more directly useful quantity ($\Delta\kappa_p$; in mm^{-1}) is the change in curvature (Birdwell et al., 2007) calculated as follows:

←

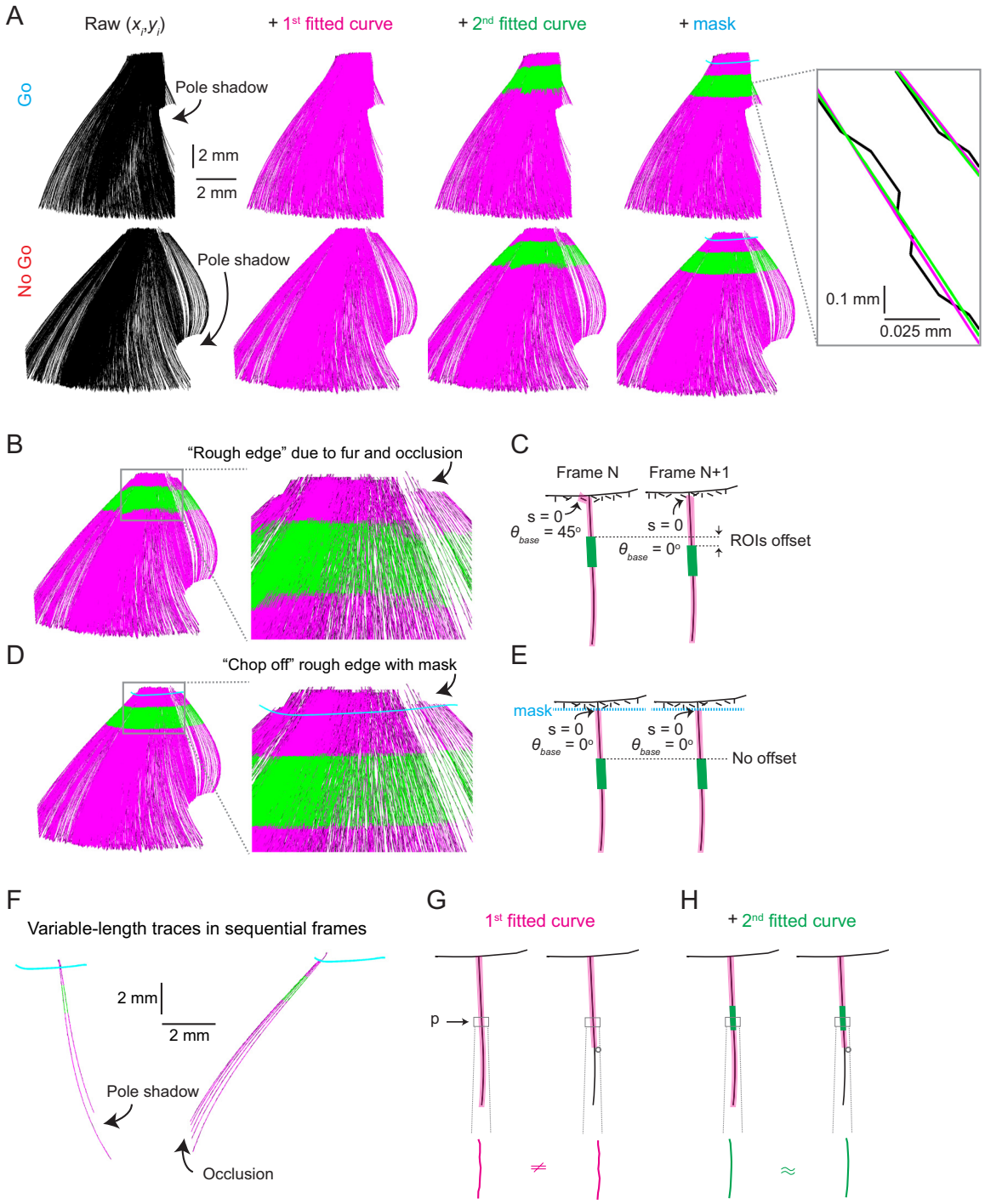

Figure 4. Schematic of video analysis methods. ***A***, Illustration of the steps involved in measuring whisker angle and curvature from raw tracked video. Each image shows a projection through time across one trial. Raw *x-y* pairs representing the medial axis of the whisker (black) are first fitted with a parametric curve (magenta, shown overlaid on black). A secondary parametric curve (green) is then fitted over an arc-length region of interest (ROI). A "mask" (blue) truncates the whisker at its noisy edge near the fur on the face and defines the arc-length origin. Magnified panel at right shows the pixilation of the raw *x-y* pairs (in black) and the smooth fitted curves (magenta and green). ***B–E***, Illustration of contamination of whisker tracing by fur on the face. Small hairs on the whisker pad can cause spurious measurements of shape and angle at the base of the whisker. ***B***, Example time-projection through a trial showing how contamination by fur and lickport occlusion can interfere with measurement of shape and angle at base, as well as interfere with defining a constant region of arc-length (note how the green arc-length ROI is staggered from frame to frame). ***C***, Schematic of problem shown in ***B*** for a single pair of frames. ***D***, Example from ***B*** improved by the addition of a mask (blue), which effectively truncates the whisker at its intersection with the mask. The mask allows only the faithfully traced portion to be used for further analysis, and prevents the arc-length ROI (green) from being staggered from frame to frame. ***E***, Schematic of the improvements shown in ***D*** for a single pair of frames. ***F–H***, Illustration of the purpose of the secondary fitted curve (green), fitted over an arc-length ROI. A parametric curve will in general have a slightly different shape when fitted over different regions of even the same whisker in the same frame. ***F***, Example of how the extent of the traced whisker in our videos could vary from frame to frame, for example due to the shadow of the pole entering the image. ***G***, Because of frame-to-frame variability in the extent of the traced whisker, the curvature of the primary fitted curve (magenta) at a point reflected not only the true whisker curvature but also fitting artifacts. Magenta shading at top indicates region of whisker fitted. Curvature is measured at the point (*p*) indicated by the arrow, and the local region around this point is magnified at the bottom. Magenta curves at bottom show the fitted curve over the local, magnified region. Left and right columns indicate a frame in which the full whisker is traced (left) and a frame in which the whisker is traced only as far as the pole (right; pole indicated by gray circle). ***H***, To minimize artifacts due to fitting variable lengths of whisker, we fitted a secondary parametric curve over a constant arc-length region of interest. Curvature measured from this secondary fitted curve was much more reliable. Green box at top indicates the arc-length region of interest over which the secondary curve was fitted. Green curves at bottom show the fitted secondary curve over the local, magnified region.

$$\Delta\kappa_p = \kappa_p - \frac{\sum_i \kappa_p(t_i)}{N}, \quad \text{(M16)}$$

where t_i is a set of N time points during which the whisker is not bent. Intrinsic curvature for the C2 whisker in the plane of the protraction was approximately 10-fold smaller than peak curvature changes and was thus ignored.

Whisker follicle position coordinates (x_f, y_f) were estimated by linearly extrapolating past the end of the tracked whisker, typically from the intersection with the mask; that is, proceeding from $c(l_{intersection})$. The extrapolation was based on θ_{base}. The distance to extrapolate (typically 1 mm) was estimated based on inspection of the video data as follows.

$$(x_f, y_f) = (-d_{extrap}\sin(\theta_{extrap}) + x(l_{base}), -d_{extrap}\cos(\theta_{extrap}) + y(l_{base})), \quad \text{(M17)}$$

where d_{extrap} is the distance to extrapolate and $\theta_{extrap} = \theta_{base}$.

Follicle position along the face (d_f) was estimated by taking the x-coordinate (anterior–posterior axis) value of follicle position, $d_f = x_f$.

The lever arm from location of curvature measurement to location of whisker-object contact, for each time point, is as follows:

$$\vec{r}_p = (x_c - x(l_p), y_c - y(l_p)). \quad \text{(M18)}$$

The angle (in radians) of this vector is as follows:

$$\theta_p = \tan^{-1}\frac{x_c - x(l_p)}{y_c - y(l_p)}. \quad \text{(M19)}$$

Similarly, the lever arm from follicle to whisker-object contact location:

$$\vec{r}_0 = (x_c - x_f, y_c - y_f). \quad \text{(M20)}$$

And the angle (in radians) of this vector is as follows:

$$\theta_0 = \tan^{-1}\frac{x_c - x_f}{y_c - y_f}. \quad \text{(M21)}$$

At each time point, we calculated the magnitude of the force applied to the whisker (F; in 1×10^{-6} N) using Equation 4. Magnitude of moment at the follicle (M_0; in 1×10^{-6} N * 1×10^{-3} m) was calculated using Equation 5. Magnitude of the axial force (F_{ax}; in 1×10^{-6} N) pushing into the follicle along the axis of the whisker was calculated using Equation 6. Magnitude of the lateral force (F_{lat}; in 1×10^{-6} N) was calculated using Equation 7.

During whisking in air, the forces acting on the follicle are dominated by the whisker's moment of inertia and its angular acceleration. These forces are much smaller than the forces produced by contact between object and whisker and not relevant to object localization. The forces were therefore set to zero between contacts (Fig. 6*B*,*D*).

We excluded trials in which the whisker contacted the pole during retraction from behavior and whisker analyses (Figs. 6, 7, 8, 9, 10). We also excluded contacts after the decision.

Simulations with friction. We used Equation M3 to compute whisker shape for a force applied at a point (x, y) along the whisker (arc-length s from the base; Fig. 5*D*–*F*). The force was of the form $F\hat{n} + \mu F\hat{t}$, where F is the force normal to the whisker, μ is the sliding friction coefficient, and $\hat{n}$,$\hat{t}$ are the unit vectors normal and tangential to the whisker, respectively. The model captures sliding friction; stick-slip events were not modeled. The force and the point along the contour where it acts were then varied until the whisker shape was consistent with contact with an object (i.e., the pole) at a particular point (methods are similar as those used in Solomon and Hartmann, 2006). A μ of 0.3 was used as a value typical for hair (Bhushan et al., 2005).

Calculating the protraction parameter. Mice protract their whiskers into objects by changing the azimuthal angle of the whisker and by moving the follicle forward. To describe protraction with a single parameter, we combined azimuthal motion and follicle translation using the protraction parameter, θ_{total} (Fig. 8*A*). At a given follicle position d_f and θ_{base}, θ_{total} is the angle through which the whisker would have to be rotated around the fulcrum for it to detach from the pole. For the frame of first touch of each contact, we defined a line with origin in the follicle and along θ_{touch}. The intersection point between this line and a line along the anterior–posterior axis passing through the approximate contact point for the Go position (5 mm from the follicle) was determined. For each video frame of a contact, the angle between the base of the whisker (i.e., a line along θ_{base}) and a line connecting the follicle to the intersection point was calculated. This angle is θ_{total}. θ_{total} at first touch is thereby by definition 0.

Statistics. Statistics were computed as indicated in the text. Comparisons of forces and bending moments for target and distracter locations (Figs. 6, 8, 9, 10) were based on bootstrap methods. For every session, we analyzed frames of contact in which θ_{total} was $\geq 2°$. We restricted the analysis to these protraction amplitudes because whisker tracking could be unreliable near the moment of whisker-pole contact (corresponding to $\theta_{total} = 0°$).

Analysis of whether the slope of $M_0(\theta_{total})$, $F_{ax}(\theta_{total})$, and $F_{lat}(\theta_{total})$ was larger for the Go trials than for No Go trials (Fig. 8*B*–*D*) proceeded as follows, separately for each of three mice.

For $M_0(\theta_{total})$, $F_{ax}(\theta_{total})$, and $F_{lat}(\theta_{total})$ time series, N_{nogo} trials were drawn with replacement 10,000 times from both the Go and No Go trials (after pooling trials across two behavioral sessions for one of the three mice), where N_{nogo} is the number of No Go trials. Therefore, each bootstrap sample comprised N_{nogo} randomly matched Go and No Go trials. For each bootstrap sample, we calculated the maximum likelihood estimate of the fraction of trial pairs in which the Go trial had a slope greater than that of the matched No Go trial (i.e., $slope_{go} > slope_{nogo}$) and the binomial 95% confidence interval for this fraction (MATLAB "binofit"). We then averaged the means and confidence intervals across all 10,000 bootstrap repetitions. For each of $M_0(\theta_{total})$, $F_{ax}(\theta_{total})$, and $F_{lat}(\theta_{total})$, and for all mice, the mean 95% confidence intervals did not contain 0.5. Specifically, for $M_0(\theta_{total})$, mean confidence intervals for the three mice were [0.59, 0.90], [0.88, 1.0], and [0.70, 0.94]; for $F_{ax}(\theta_{total})$, they were [0.58, 0.90], [0.85, 1.0], and [0.71, 0.95]; and for $F_{lat}(\theta_{total})$, they were [0.60, 0.91], [0.88, 1.0], and [0.71, 0.95]. Slopes were thus larger for Go trials at the $\alpha = 0.05$ significance level.

Analysis of which force/moment cues could underlie performance on the illusion experiments (Fig. 9, Fig. 10) proceeded similarly, separately for each of two mice. For $M_0(\theta_{total})$, $F_{ax}(\theta_{total})$, and $F_{lat}(\theta_{total})$ time series, $N_{illusion}$ trials were drawn with replacement 10,000 times from both the illusion and No Go trials (after pooling trials across three behavioral sessions for each mouse), where $N_{illusion}$ is the number of illusion trials. Therefore, each bootstrap sample comprised $N_{illusion}$ randomly matched illusion and No Go trials. By bootstrapping in this way, we avoided the problem of having to decide which particular trials to compare between illusion and No Go trial types (because the number of trials of each type was not matched and there was no logical way to pair them). We restricted analysis to No Go trials in which the pole was in one of the two locations closest to the Go location. The rationale for choosing locations close to the Go location is that to achieve its level of illusion-trial task performance, the mouse must have been able to discriminate illusion trials (in which a flexible object was in the Go location) from even the most similar No Go trials. On the other hand, we chose the two closest No Go positions, rather than only the closest, in order to have more No Go trials for analysis (there was no obvious dependence of performance on the particular No Go location). For each bootstrap sample, we calculated the maximum likelihood estimate of the fraction of trial pairs in which the illusion trial had a slope greater than that of the matched No Go trial (i.e., $slope_{illusion} > slope_{nogo}$) and the binomial 95% confidence interval for this fraction. We then averaged the means and confidence intervals across all 10,000 bootstrap repetitions. For analysis of $F_{ax}(M_0)$, the procedure was identical except that we estimated the fraction of trial pairs in which the illusion trial had a slope less than that of the matched No Go trial (i.e., $slope_{illusion} < slope_{nogo}$).

For the data shown in Figure 6*G*–*I*, we investigated whether performance depended on absolute force amplitudes. Calculations were done separately for each trial type of each session. We fitted a generalized linear model for binomial data with a logit link function (MATLAB "glmfit") to force versus response. The reported p values are for the null hypothesis that response does not depend on force.

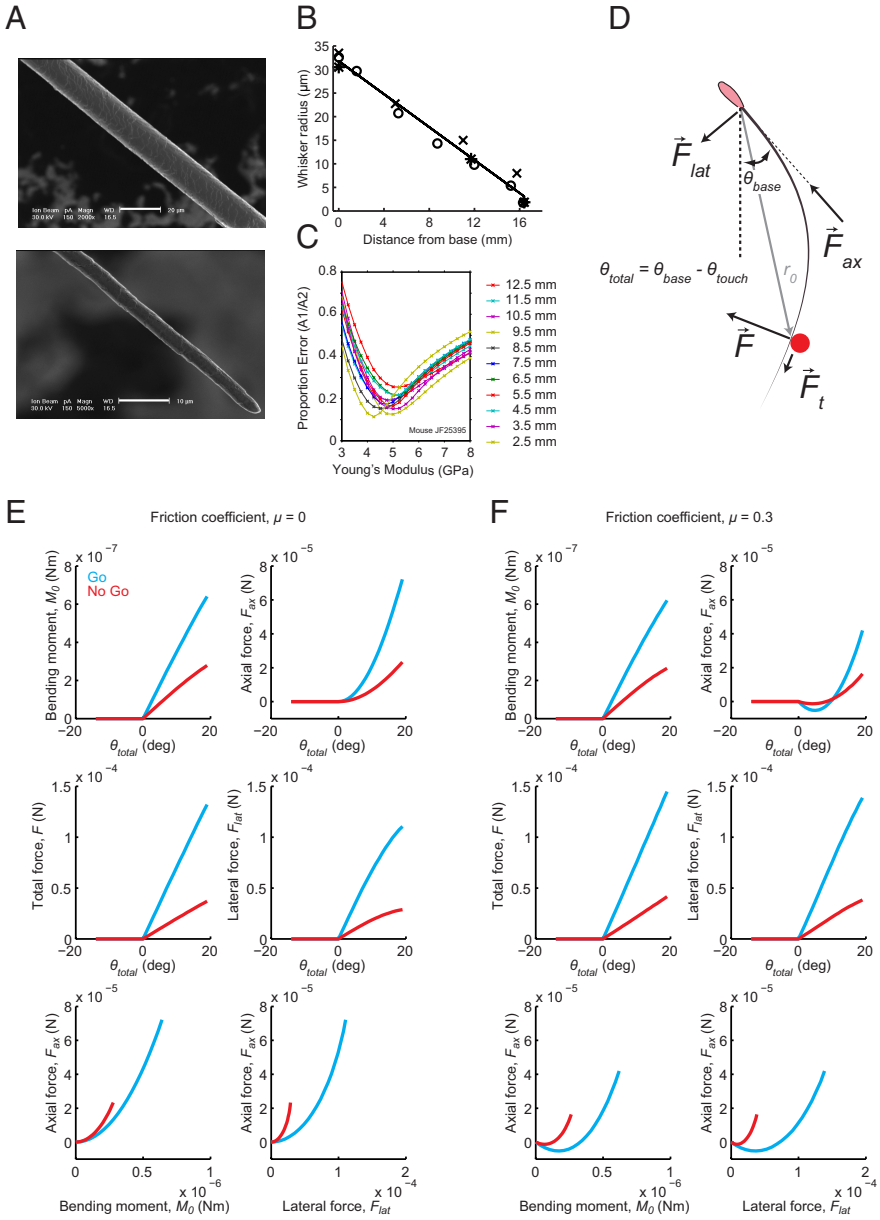

Figure 5. Structure of C2 whiskers and simulations without and with sliding friction. ***A***, Scanning electron microscopy images of one C2 whisker toward the middle of the whisker (top: scale bar, 20 μm) and the tip of the whisker (bottom: scale bar, 10 μm). ***B***, Radii of three C2 whiskers (*x* and *, light microscopy; o, electron microscopy). ***C***, Results of Young's Modulus estimation for one C2 whisker. Comparison of measured values and model predictions for deflections of a C2 whisker by force application at 1 mm increments along the whisker (different curves, indicated by legend). Model prediction error (*y*-axis) was calculated for a range of Young's Modulus values (*x*-axis). Values of Young's Modulus that minimized the error between model and measurement clustered around 5 GPa. See Materials and Methods for details. ***D***, Schematic of the simulation (***E***, ***F***). A conical whisker presses against a stiff pole. Forces act both normal to the whisker and tangential to the whisker (if the friction coefficient is not zero). The axial force pushes the whisker into the follicle, whereas the lateral force pushes the whisker laterally within the follicle. Friction results in a force pulling the whisker out of the follicle (F_t). Lateral stresses in the follicle are dominated by the bending moment on the whisker. Here, $\theta_{total} = \theta_{base} - \theta_{touch}$ because protractions were not associated with follicle translation (compare with Fig. 8*A*). ***E***, Simulated forces and bending moments for friction coefficient $\mu = 0$, so that the pole exerts a force normal to the whisker ($F_t = 0$). Target, 5 mm; distracter, 8 mm. ***F***, Simulated forces and bending moments for friction coefficient $\mu = 0.3$, so that $F_t = 0.3\,F$. In the presence of friction, forces are not normal to the whisker. Furthermore, the axial force changes sign as a function of protraction. For small protractions, the whisker is pulled out of the follicle, whereas for larger protractions, the axial force pushes the whisker into the follicle. Target, 5 mm; distracter, 8 mm.

For decoding lick/no-lick behavioral choice using linear discriminant analysis (MATLAB "classify"), we pooled video time points across trials in each illusion behavioral session. We resampled illusion trials randomly with replacement such that they occurred at the same frequency as the normal (stiff-pole) Go trials. We then used linear discriminant analysis to predict the class membership (coming from a lick or a no-lick trial) for each time point for the indicated two-dimensional time series ($\{F_{ax}, M_0\}$, $\{M_0, \theta_{total}\}$, $\{F_{ax}, \theta_{total}\}$, or $\{F_{lat}, \theta_{total}\}$). Decoder performance was quantified as the fraction of time points correctly assigned as coming from a lick or a no-lick trial.

Glossary

a_{base}: Radius of whisker cross section at the base

a_p: Radius of whisker cross section at p

$c(l)$: Parametric curve fitted to sequence of (x_i, y_i) pairs returned by tracking

$\hat{c}(l)$: Parametric curve fitted to sequence of (x_i, y_i) pairs from tracker, within an arc-length region of interest

d_f: Position of the follicle along the face

d_{pole}: Distance between whisker and pole

E: Young's Modulus

$\vec{F}$: Force exerted on the whisker by the object

$\vec{F}_{ax}$: Component of F pushing the whisker into the follicle

$\vec{F}_{lat}$: Component of F pushing the whisker against the caudal wall of the follicle

F_{scale}: Force acting on the scale when pushing the whisker against it

$\vec{F}_t$: Friction-caused force pulling the whisker out of the follicle

I_p: Second moment of inertia at p

$\vec{M}_0$: Bending moment acting on the follicle

$\vec{M}_p$: Bending moment acting on p

p: Point along the whisker

$\vec{r}_0$: Vector (lever arm) from the follicle to the point of contact

$\vec{r}_p$: Vector (lever arm) from p to the point of contact

s: Arc-length distance along whisker

s_p: Length of the whisker along its contour from the base to point p

x_c: Image x-coordinate of whisker-object contact point

x_f: Image x-coordinate of whisker follicle

y_c: Image y-coordinate of whisker-object contact point

y_f: Image y-coordinate of whisker follicle

θ: Angle of whisker, defined at all points along whisker

θ_0: Angle of $\vec{r}_0$

θ_{base}: Angle at the base of the whisker

$\theta_{contact}$: Whisker angle at the point of contact between object and whisker

θ_p: Angle of $\vec{r}_p$

θ_{total}: Parameter quantifying protraction; combines angular rotation and translation

θ_{touch}: θ_{base} at the time of first touch during a contact period

κ_p: Curvature of the whisker at p

$\Delta\kappa_p$: Change in curvature of the whisker at p

μ: Friction coefficient

Results

Head-fixed mice perform radial object localization

We trained head-fixed mice in one of several variations of a Go/No Go vibrissa-based object localization task (Fig. 2*A*–*C*). Mice reported the presence of a vertical pole within a target position (the "Go stimulus"; proximal) or in a distracter position (the "No Go stimulus"; distal) by either licking (Go response) or

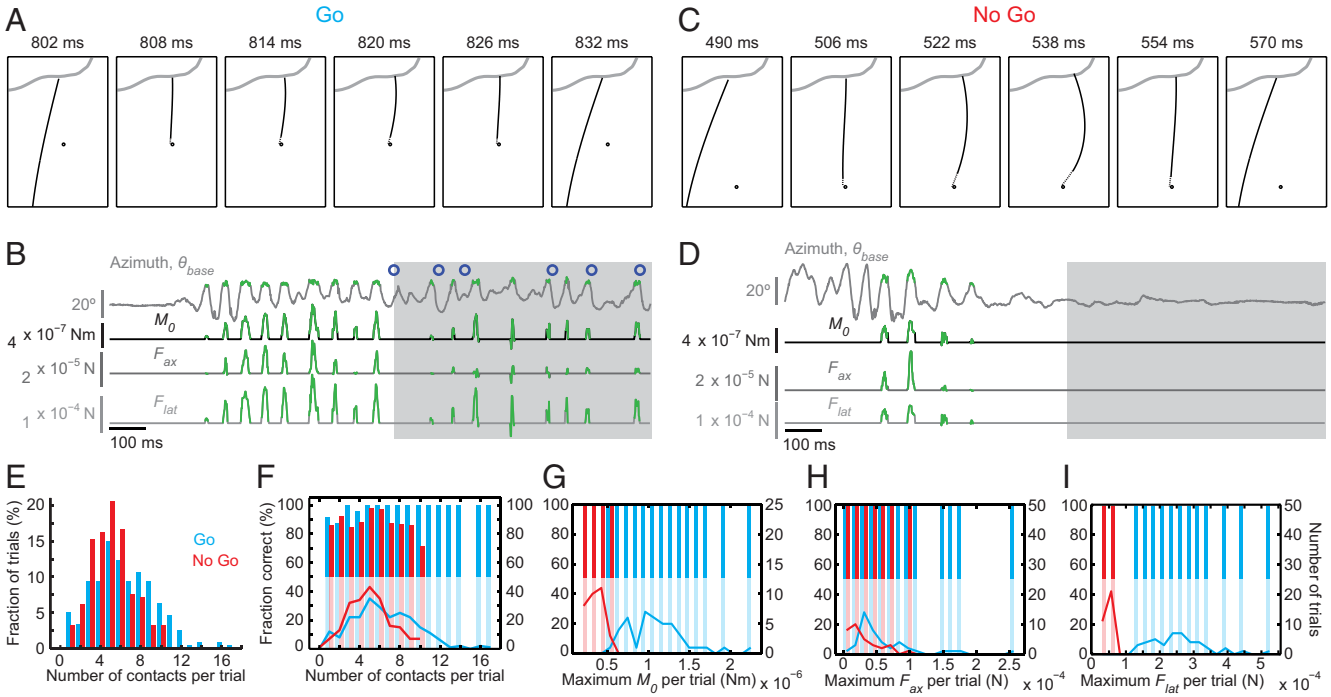

Figure 6. Whisking and contact forces underlying radial object location. ***A***, Time-lapse sequence showing one contact. The black line is the medial axis of the tracked whisker. The small circle indicates the cross section of the object (a thin pole). Distance from the pole to the follicle, 5 mm. ***B***, θ_{base}, M_0, F_{ax}, and F_{lat} for one correct Go trial. Green indicates manually scored contact periods. The initial time point in a contact period is defined as the first touch (first video frame showing contact between whisker and object). Blue circles indicate licks. The first answer lick defines the reaction time. Gray patch is after reaction time. ***C***, ***D***, Same as ***A*** and ***B*** but for a correct No Go trial. Distance from the pole to the follicle, 8 mm. ***E***, The distribution of the number of contacts for Go (blue) and No Go (red) trials before the reaction time (3 mice, 5 sessions). ***F***, The fraction of trials correct as a function of the number of contacts for Go (blue) and No Go (red) trials (three mice, five sessions). Bars indicate performance (left *y*-axis); lines, number of trials with a given number of contacts (right *y*-axis). ***G***–***I***, Moments and forces across trial types. Performance as a function of the maximum value of a mechanical parameter across trials for Go (blue) and No Go (red) trials (one mouse, one session). Bars indicate performance (left *y*-axis); lines, number of trials with a particular maximum moment or force (right *y*-axis). Note that this mouse performed perfectly for the 100 consecutive trials shown. ***G***, Maximum bending moment. ***H***, Maximum axial force. ***I***, Maximum lateral force.

withholding licking (No Go response). In each trial, the pole was presented at a single location and the mouse had to judge object location based on its memory of the positions. Our task probes memory-guided object localization in a head-centered reference frame (O'Connor et al., 2010a). Whiskers were trimmed so that mice performed the task with a single whisker (C2). Target and distracter locations were carefully arranged for each mouse along the "radial" axis. Therefore, whiskers contacted the pole in both locations at nearly identical azimuthal angles (angle difference: mean 1.6°; range 0.2–2.9°; 3 mice, 5 sessions; Fig. 2*B*). The remaining azimuthal differences are likely too small to be discriminated (O'Connor et al., 2010a; see also below).

All mice (8/8) reached a 90% correct response criterion on the standard task (single target and distracter locations; distance from the follicle: target, 5 mm; distracter, 8 mm) over hundreds of trials per session (Fig. 2*D*). To estimate reaction times, we measured the time elapsed between the pole coming within reach and the first (answer) lick on Hit trials (Fig. 2*C*). These reaction times were ~640 ± 79 ms (mean ± SD, 3 mice), longer than for object localization in the anterior–posterior axis with a single whisker (~470 ± 210 ms; A.H., D.H.O., and K.S., unpublished data). This difference suggests that radial object localization requires more extensive interactions between whisker and object compared with anterior–posterior localization.

We next explored the accuracy of vibrissa-based radial object localization. Psychometric curves relate performance to the size of the offset between target and distracter locations. All mice generalized from the standard task to smaller distances between distracter and target stimuli (target, 5 mm; distracter, 6–9 mm; Fig. 2*E*), and performed above chance even with target and distracter locations separated by one millimeter (mean fraction correct = 57%, all mice $p < 5 \times 10^{-3}$, one-sided binomial test; Fig. 2*F*).

Computing the forces acting on the follicle

We used high-speed videography (O'Connor et al., 2010a) to determine the mouse's motor strategies, and also to measure the mechanical forces acting on the whiskers that must support radial object localization. Automated whisker tracking provided the contour of the whisker. To account for imperfections in videography and whisker tracking (i.e., pixilation, noisy tracking close to the face due to fur, limited field of view, failure of tracking near the pole due to shading), we developed a number of practical procedures to enable force estimates from the video data (Fig. 3*A*–*E*; see Materials and Methods). Changes in the whisker curvature in the plane of whisking were used to estimate forces acting on the follicle (Fig. 3*F*–*I*).

Whiskers are cantilevered beams, with one end embedded in the follicle in the whisker pad (Fig. 1*A*). As the whisker protracts against the pole, the pole exerts a force on the whisker ($\vec{F}$), bending it and causing stresses in the follicle (Fig. 1*B*). With negligible friction, $\vec{F}$ will act in a direction normal to the whisker at the point of contact. We can calculate the relationship between whisker shape and the force on the whisker in the quasi-static regime. This means that we ignore small force transients on millisecond time scales (Den Hartog, 1947; Timoshenko et al., 1974), which under our conditions had amplitudes <10% of peak forces and lasted ~1 ms (A. Efros, S.A. Hires, and K. Svoboda, manuscript in

preparation). We also assume that the C2 whisker is contained in the plane of whisker movement (Knutsen et al., 2008; Quist and Hartmann, 2012). Because the whisker is a long lever arm, lateral stresses in the face (i.e., perpendicular to the whisker) are dominated by the bending moment ($\vec{M}_0$) acting on the follicle (Birdwell et al., 2007) are calculated as follows:

$$\vec{M}_0 = \vec{r}_0 \times \vec{F}, \quad (1)$$

where $\vec{r}_0$ is the lever arm connecting the whisker base to the point of contact. $\vec{M}_0$ rotates the follicle in the horizontal plane around a fulcrum in the whisker pad. For thin elastic beams, such as whiskers (Fig. 5), the bending moment acting on a point (p) along the contour of the beam ($\vec{M}_p$) is proportional to the change in curvature ($\Delta\kappa_p$) at p (Birdwell et al., 2007; Fig. 3*F*,*G*). The constants of proportionality are governed by the shape and material properties of the beam (Landau and Lifshitz, 1986). Coefficients of friction for hair are generally low (0.1–0.3; Bhushan et al., 2005). We therefore neglect friction in the subsequent calculations. Numerical simulations show that our conclusions are expected to be qualitatively similar with sliding friction taken into account (Fig. 5*D*–*F*). With the help of standard trigonometric identities one can arrive at the expression:

$$M_p = r_p F \cos(\theta_p - \theta_{contact}) = (\Delta\kappa_p)(EI_p), \quad (2)$$

where r_p is the length of the lever connecting p and the point of contact with the pole, θ_p is the angle of the lever, and $\theta_{contact}$ is the angle of the whisker at the contact point and determines the direction of the applied force. (EI_p) is the bending stiffness of the whisker at p. E is the Young's Modulus, which characterizes the elasticity of the whisker. I_p, the second moment of inertia at p, is a purely geometric quantity:

$$I_p = \frac{\pi a_p^4}{4}, \quad (3)$$

where a_p is the whisker radius at p. We measured E and a_p for whiskers used by mice during the experiment (Fig. 5; see Materials and Methods). The time-dependent $\Delta\kappa_p$ was calculated for each whisker video frame. Equations 2 and 3 then allow us to extract the magnitude of $\vec{F}$ from measured quantities as follows:

$$F = \frac{(\Delta\kappa_p)(EI_p)}{r_p \cos(\theta_p - \theta_{contact})}. \quad (4)$$

The bending moment at the follicle ($\vec{M}_0$) has magnitude:

$$M_0 = r_0 F \cos(\theta_0 - \theta_{contact}), \quad (5)$$

where θ_0 is the angle of the lever arm (Fig. 3*H*). $\vec{M}_0$ primarily causes lateral stresses in the skin. In addition, the axial component of the force, $\vec{F}_{ax}$, causes axial stresses by pushing the whisker into the follicle (Fig. 3*I*), with a magnitude:

$$F_{ax} = F \sin(\theta_{base} - \theta_{contact}), \quad (6)$$

where θ_{base} is the angle of the whisker at the base, and reflects the orientation of the follicle.

The lateral component of the force, $\vec{F}_{lat} = \vec{F} - \vec{F}_{ax}$, pushes the whisker against the posterior side of the follicle when protracting against an object, with magnitude:

$$F_{lat} = F \cos(\theta_{base} - \theta_{contact}). \quad (7)$$

The order of calculations is illustrated in Figure 3*F*–*I*.

Whisking and forces during radial object localization

In the radial distance discrimination task, mice were forced to whisk on every trial against the pole to determine its location. The peak whisking (θ_{base}) amplitude during the sampling period was 24 ± 6° (mean ± SD over trials; 3 mice, 4 sessions, 355 trials). Touch forces were quantified in a subset of sessions (3 mice, 5 sessions, 445 trials), for which contact between whisker and pole was scored manually in individual image frames (Fig. 6*A*–*D*). On correct Go trials, whiskers contacted the pole on average 7.0 ± 2.0 (mean ± SD) times (range = 1–17) before the mouse indicated its decision with licking (Fig. 6*B*,*E*). On No Go trials, the rate of contacts was similar, but the number of contacts was smaller (average, 5.4 ± 1.3; range 1–10; unpaired t test $p < 10^{-3}$; Fig. 6*D*,*E*). Behavioral performance was almost equally reliable for trials with small numbers of contacts as for trials with large numbers of contacts (Fig. 6*F*). These observations suggest that mice palpate the object to accumulate evidence until reaching a decision with a certain level of confidence.

Peak moments and forces were on the order of 1 μNm and 100 μN, respectively (Fig. 6*G*–*I*). The bending moments and forces also differed across trial types. On average, peak bending moments and forces were larger for the proximal Go location compared with the distal No Go location (F_{ax}: Go = 36 ± 28 μN, No Go = 19 ± 19 μN; F_{lat}: Go = 210 ± 69 μN, No Go = 39 ± 14 μN; M_0: Go = 0.9 ± 0.3 μNm, No Go = 0.3 ± 0.1 μNm). This is largely due to the linear taper of rodent whiskers (Birdwell et al., 2007; Fig. 5; $r = -0.988$), which implies that the bending stiffness decreases steeply with distance from the follicle (Eq. 3). M_0, F_{ax}, and F_{lat} therefore depend on object location: during a protraction against an object, bending moments and forces build up more rapidly for more proximal objects (Fig. 5*E*).

In addition to pivoting whiskers around a fulcrum within the follicles, causing changes in the azimuthal angle θ_{base} of the whisker, mice also move the follicle (d_f) horizontally along the face (Figs. 2*B*, 7*A*). θ_{base} and d_f are typically correlated, but they can also move independently (Harvey et al., 2001; O'Connor et al., 2010a). Indeed, follicles moved gradually forward over multiple whisker-object contacts within a trial; as a result, θ_{base} at the time of the first touch within each contact period (θ_{touch}; measured from the first video frame scored as occurring during a touch event) decreased across successive contacts (Fig. 7*A*). As a simple consequence of geometry (Fig. 7*B*), these changes in θ_{touch} are always larger for more proximal object locations than for more distal locations, providing possible azimuthal cues for radial object location (Figs. 1*D*, 7*C*).

In our experiments, mice might therefore avoid force-dependent radial object localization by relying on these azimuthal cues. Specifically, mice could theoretically track θ_{touch} (or a related parameter, such as θ_{base} averaged over one contact period) and d_f to extract object location (as illustrated in Fig. 1*D*). Two observations argue against this possibility. First, we analyzed whether task performance depended on the relationship between θ_{touch} and d_f over multiple contacts in individual trials. Mice generally performed at high levels in selected trials with relatively constant θ_{touch} (range of $\theta_{touch} < 1°$, performance = 85% correct, $n = 39$). Even on trials with single contacts, mice performed radial object localization (performance = 84% correct, $n = 19$; Fig. 7*D*). Second, we randomly varied the pole position in the azimuthal dimension (±2–3°) while keeping the radial distance constant (Fig. 7*E*). This azimuthal jitter interfered with the object-location-dependent relationship between θ_{touch} and d_f (Fig. 7, compare *C*, *F*), but did not perturb performance of the

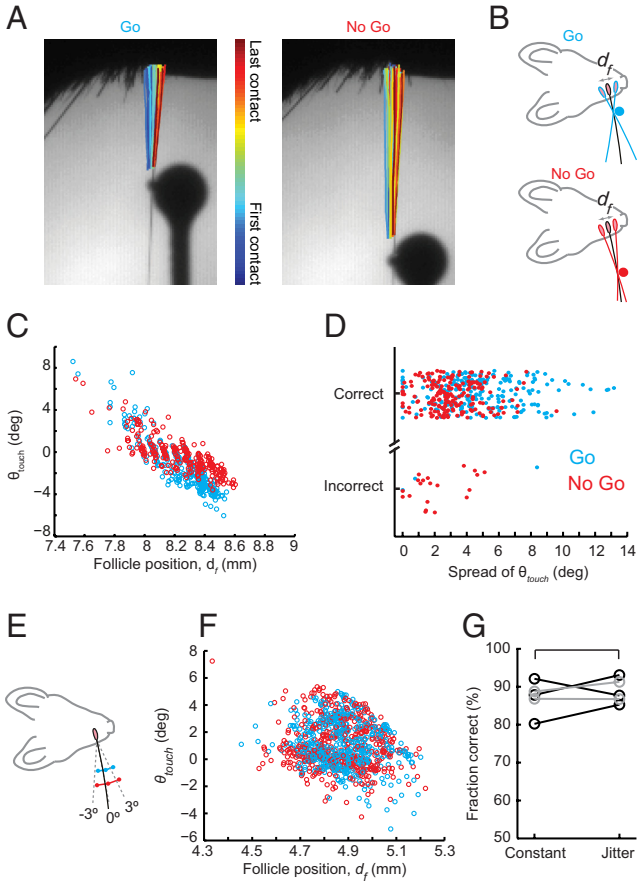

Figure 7. Mice rely on stresses in the follicle for object localization. ***A***, Spread of the azimuthal angle at first touch (θ_{touch}) across contacts of one session. Later (earlier) contacts within trials are coded by warmer (colder) colors. Distance from the pole to the follicle: Left, 5 mm, Go; right, 8 mm, No Go. Note that the spread of θ_{touch} is larger for proximal objects (left) compared with distal objects (right), producing a possible azimuthal cue for object localization. ***B***, The larger spread of θ_{touch} with changing d_f for more proximal object locations is a consequence of geometry. ***C***, Azimuthal angle at first touch (θ_{touch}) as a function of whisker pad movement, d_f, for one session. Difference in the slope of Go (blue) versus No Go (red) data points represents greater impact of d_f on θ_{touch} for more proximal objects (corresponding to Go). Target, 5 mm; distracter, 8 mm. Same session as in ***B***. Striped pattern reflects finite resolution of whisker tracking due to fixed pattern noise in video images. ***D***, Correct and incorrect trials as a function of the spread of θ_{touch} across contacts (three mice, five sessions). Each point corresponds to a trial. Points are randomly displaced in the *y*-axis for clarity. Values of zero correspond to trials with single contacts. ***E***, Schematic showing the introduction of 3° of azimuthal jitter. See Materials and Methods for details. ***F***, θ_{touch} as a function of d_f with azimuthal jitter (±3°). Target, 5 mm; distracter, 8 mm (one mouse, one session). The slopes of Go (blue) versus No Go (red) points are indistinguishable. ***G***, Performance on sessions without (first 50 trials) and with (rest of the session, >100 trials) random azimuthal jitter (two mice shown by black and gray data points, five sessions). Azimuthal jitter consisted of azimuthal angle offsets (range ± 3°) added to pole positions across trials.

radial distance task (Fig. 7*G*). Performance was not reduced even in the first session in which jitter was introduced, excluding the possibility that mice relearned the radial task after introduction of jitter. Azimuthal jitter in the pole position also introduced uncertainty in the timing of contact events within a protraction (Go trials, ~4.9 ms; No Go trials, ~6.2 ms; calculated based on the mean velocity at contact and ±3° jitter). Radial distance discrimination in the presence of azimuthal jitter also shows that azimuthal differences between the pole positions do not serve as cues for discrimination (O'Connor et al., 2010a; see also below).

The pneumatic sliders moving the pole vertically into reach of the whisker could cause distance-dependent vibrational cues. We performed control experiments as follows. During the first session, behavioral training was under standard conditions (Go, 5 mm; No Go, 8 mm). For the next two sessions, the pole was moved smoothly from an anterior out-of-reach position into the target (5 mm) or distracter (8 mm) locations in the horizontal plane. In these sessions, without sliders, performance was not reduced (fraction correct: control = 83.4%, n = 543; without sliders = 85.0%, n = 853; two-sided binomial test, p = 0.23), arguing against vibrational cues. In addition, radial object localization is whisker dependent. In three highly performing mice, we cut the C2 whisker. After trimming, performance dropped to chance levels (fraction correct = 50.4%; one-sided binomial test, p = 0.28). These data suggest that mice perform force-dependent radial object localization based on measurement of the time-varying forces in the follicle.

Force cues underlying object localization

We next investigated the variables that could underlie force-dependent radial object localization. M_0, F_{ax}, and F_{lat} all increased with protraction against the object, and the rates of change depended on object location (Fig. 8). Whisker deformation by the object is caused by changes in whisker angle (θ_{base}) as well as translation of the follicle along the face (d_f; Figs. 2*B*, 7*A*). We therefore parameterized protraction by the protraction parameter θ_{total} (Fig. 8*A*, see Materials and Methods). $M_0(\theta_{total})$, $F_{ax}(\theta_{total})$, and $F_{lat}(\theta_{total})$ all had larger slopes for the more proximal target location compared with the distracter location (M_0, $p < 0.05$; F_{ax}, $p < 0.05$; F_{lat}, $p < 0.05$; see Materials and Methods; Fig. 8*B–D*). In the presence of sliding friction, F_{ax} would pull on the whisker for small protractions and then push the whisker into the face at larger protractions (Fig. 5*D–F*).

Because the durations of contacts were similar for both object locations (Go, 23.2 ± 13.8 ms; No Go, 20.3 ± 12 ms; mean ± SD; 1690 contacts; p = 0.054, unpaired two-sample t test), the time-derivatives of the moment and force variables, which are likely more directly related to driving sensory responses, are also object location dependent. Therefore, mice could use the object location-dependent variables M_0, F_{ax}, and F_{lat} (Fig. 8*B–D*), together with knowledge about whisker movement, to compute radial object location.

Do changes in individual mechanical variables as a function of protraction code for object location? We tested this possibility by confounding the relationships between $M_0(\theta_{total})$, $F_{ax}(\theta_{total})$, and $F_{lat}(\theta_{total})$ and object location (Fig. 9). Mice were trained to lick when the object was present in a proximal target location and had to withhold licking when the object was in a distal distracter location. For every No Go trial, the distracter location was chosen randomly from the range 7–13 mm to produce a range of forces across different No Go trials. Within bins of 32 consecutive trials, two randomly chosen Go trials were not rewarded. In half of these trials, a compliant object (a rat's whisker) was presented in the target location with the stiff pole out of reach ("illusion" trials; Fig. 9*A*). In all other trials, the compliant object was out of reach. Before every session, the position of the compliant object (for illusion trials) was matched to the Go position of the stiff object based on video still frames. Differences in θ_{touch} were similar between contacts with the stiff and compliant objects in the target position (2 mice, 6 sessions, mean difference = 0.9°; range = 0.1–2°) and stiff object in the target and distracter positions (2 mice, 6 sessions, mean difference = 1.5°; range = 0.3–2.9°). In this type of experiment, each session thus contained three trial types (Go, stiff; Go, compliant; No Go, stiff). In addi-

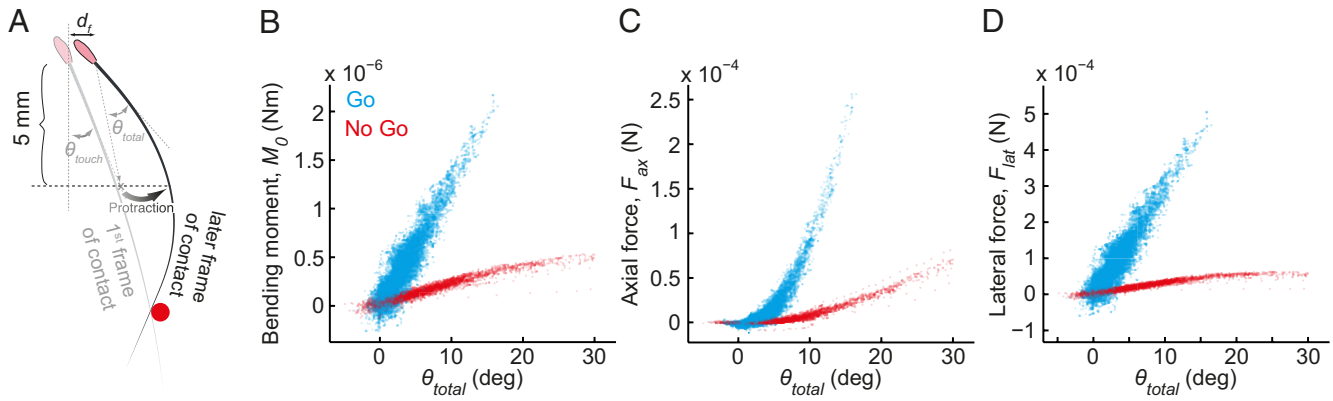

Figure 8. Stresses on the follicle during object localization. ***A***, Schematic illustrating computation of the protraction parameter θ_{total}. θ_{total} accounts for protraction due to changes in both θ_{base} and d_f. The same whisker is shown early (gray) and late (black) during a protraction. Gray "x" marks the intersection point of a line defined by θ_{touch} with a line running along the anterior–posterior axis at the approximate lateral distance of the contact point for the Go position (5 mm from the follicle). ***B–D***, Bending moment (***B***), axial force (***C***), or lateral force (***D***), acting on the follicle as a function of the protraction parameter for Go (blue) and No Go (red) trials. All time points corresponding to contact periods in one session are overlaid with partial transparency.

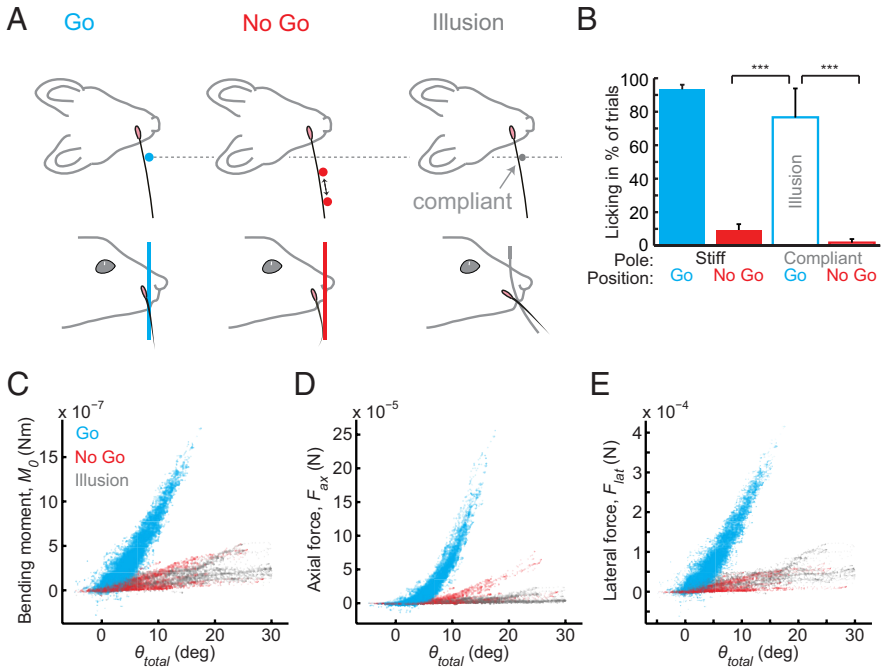

Figure 9. The amplitudes of bending moment and axial force are not the primary cues underlying object localization. ***A***, Schematic of the "illusion" experiment. Mice discriminated a proximal Go position (5 mm) and a range of distal No Go positions (7–13 mm). In a subset of unrewarded illusion trials, a compliant pole replaced the stiff pole in the Go position. ***B***, Behavior for trials with stiff and compliant poles (five mice, 36 sessions). The fraction of trials with licks during the answer period is plotted for different conditions. ***C–E***, Parameters producing stresses in the follicle for Go trials with the stiff pole (blue), No Go trials with the stiff pole (red), and unrewarded Go trials with the compliant pole (gray; one session). ***C***, M_0 as a function of θ_{total}. ***D***, F_{ax} as a function of protraction. ***E***, F_{lat} as a function of protraction.

tional experiments, the compliant pole was presented in the middle of the distracter position range (No Go, compliant; 4 mice, 8 sessions; Fig. 9*B*).

As the whisker protracts against the compliant object, bending moments and forces acting on the follicle build up more slowly compared with the stiff pole (compare slopes of "Go" and "Illusion" data in Fig. 9*C–E*). Object locations and the stiffness of the compliant object were chosen so that the bending moment and forces experienced in these illusion trials were in the range of those experienced in No Go trials. $M_0(\theta_{total})$, $F_{ax}(\theta_{total})$, and $F_{lat}(\theta_{total})$ could not be distinguished between cases in which the compliant object was in the target location or the stiff object was in the distracter location (Fig. 9*C–E*, see Materials and Methods).

Therefore, if $M_0(\theta_{total})$, $F_{ax}(\theta_{total})$, and $F_{lat}(\theta_{total})$ by themselves were the mechanical variables underlying object localization, then mice would group the compliant object in the target location with the stiff object in the distracter location, causing a dramatic drop in behavioral performance; mice would incorrectly make No Go responses even though the compliant object was in the Go position (and therefore be fooled into misinterpreting the location of the compliant object; a somatosensory illusion).

Remarkably, mice interpreted the compliant object in the target location as a Go stimulus (percentage of trials with licking: illusion, 77 ± 17%; control, 94 ± 3%; mean ± SD; unpaired *t* test, $p = 0.06$; Fig. 9*B*), despite the fact that the mechanical variables at a given protraction were comparable to those produced by the stiff pole in the distracter (No Go) location (Fig. 9*C–E*). Similarly, when the compliant object was presented in the distracter location, it was correctly interpreted as a No Go stimulus (illusion, 2 ± 2%; control, 9 ± 4%), excluding the possibility that mice interpreted the compliant object as a novel stimulus triggering licking (Fig. 9*B*).

These data exclude the possibility that $M_0(\theta_{total})$ and $F_{ax}(\theta_{total})$ are the solitary variables underlying object location. The situation is less clear for F_{lat}. The slope of $F_{lat}(\theta_{total})$was slightly larger for illusion trials compared with No Go trials on average, although this difference was not significant at our level of statistical power ($p > 0.05$; see Materials and Methods). However, we cannot rule out that mice perform more precise measurements of F_{lat} than our apparatus, based on which they could group the compliant object in the target location with the stiff object in the target location based on $F_{lat}(\theta_{total})$ alone (Figs. 9*E*, 10*A*). We note how-

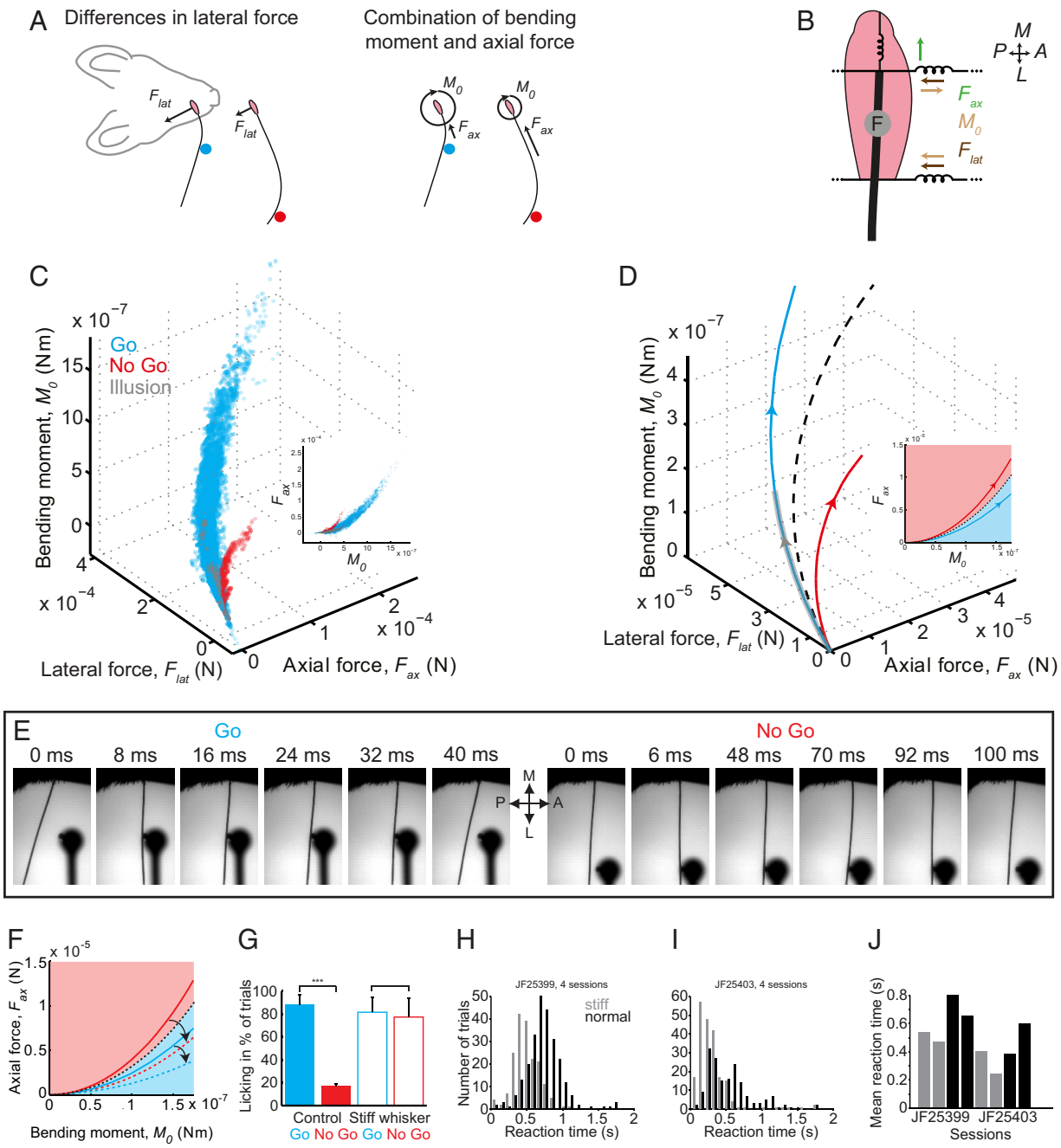

Figure 10. Force-dependent schemes for object localization. ***A***, Forces and moments at the follicle that could be read out for force-dependent radial object localization. Left, F_{lat}. Right, Mice could use a combination of at least two of the three force/moment variables, with M_0 and F_{ax} being the most likely. ***B***, Schematic illustrating how forces and the bending moment might couple to different directional stresses in the follicle. The whisker (thick black line) is embedded in the follicle (pink). It is coupled to the whisker pad through springs and rotates around the fulcrum (F). A indicates anterior; P, posterior; M, medial; and L, lateral. The bending moment and lateral force mainly cause compression or rarefaction in springs along the A-P axis, whereas the axial force affects the spring along the M-L axis. Note that little is known about stresses in the follicle that are actually sensed by mechanosensitive cells. ***C***, Forces and bending moment in Go trials with the stiff pole (blue), No Go trials with the stiff pole (red), and Go trials with the compliant pole (gray; same session as Fig. 9*D–F*). Data from Go trials with the compliant pole (gray) is masked by the data from Go trials with the stiff pole (blue). Inset: The axial force as a function of bending moment for Go trials with the stiff pole, No Go trials with the stiff pole, and unrewarded Go trials with the compliant pole (same session as Fig. 9*D–F*). ***D***, Simulated forces and bending moment in Go trials with the stiff pole (blue), with the compliant pole (gray) and No Go trials with the stiff pole (red). The dashed black line midway between Go and No Go indicates a putative decision boundary. Arrowheads indicate direction along the trajectories when protracting against an object. Inset: Projection in the F_{ax}–M_0 plane. The decision boundary separates distal objects and proximal objects in our binary choice task. ***E–J***, Whisker-stiffening experiment. ***E***, Time-lapse images of whisker-object contacts for each trial category in the whisker stiffening experiment. Note the small curvature changes of the whisker upon contacting the object. Target, 5 mm; distracter, 8 mm. ***F***, Model underlying the stiffened whisker experiment. The solid lines show simulated F_{ax} as a function of M_0 for Go (blue) and No Go (red) positions. Mice could measure F_{ax} and M_0 and judge all points above a decision boundary (dotted black line) as corresponding to a distal object, and below the boundary as corresponding to a proximal object. Stiffening the whisker reduces axial force for a given bending moment (black arrows and dashed lines), dropping the red curve below the decision boundary. The model predicts that mice with stiff whiskers will misinterpret distal objects as a proximal object. ***G***, Behavior for trials with stiff and control whiskers (two mice, four consecutive sessions per mouse). Before every session (distance from the follicle, Go 5 mm, No Go 8 mm) three cut whiskers from the contralateral side were glued to the principal whisker with hairspray, with two of them antiparallel and the third parallel (see "Whisker trimming and manipulation" section). This stiffened the whisker and abolished its taper. F_{ax} was therefore small compared with all values of M_0. Distal objects now produced combinations of M_0 and F_{ax} that were usually only seen for proximal objects. The fraction of trials with licks during the answer period is plotted for different conditions. Mice now interpreted distal objects as proximal objects (unpaired *t* test, $p = 0.8$). The behavioral responses are (*Figure legend continues.*)

ever that M_0 and F_{lat} both produce lateral stresses in the follicle (Fig. 10*A*,*B*), where M_0 is expected to dominate because of mechanical advantage (Eqs. 1, 7). It therefore seems unlikely that F_{lat} could be sensed independently of M_0 to provide an estimate of object location.

Our findings are consistent with two mechanical models for radial distance perception (Fig. 10*A*). The mouse could monitor lateral force (F_{lat}) alone (Fig. 10*A*, left) or could compare at least two variables in the three-dimensional force/moment space (Fig. 10*A*, right). Because F_{ax} produces stresses primarily in a separate part of the follicle (toward the medial end of the follicle) from M_0 and F_{lat} (Fig. 10*B*), and because the lateral stresses are dominated by M_0 (Eqs. 1, 7), the most likely possibility is a combination of M_0 and F_{ax} coding for radial object location (Fig. 10*C*,*D*). Indeed, we found that a simple decoder based on linear discriminant analysis of reported distance performed best using a combination of M_0 and F_{ax} compared with any of M_0, F_{ax}, or F_{lat} alone [specifically, $F_{ax}(M_0)$ compared with $M_0(\theta_{total})$, $F_{ax}(\theta_{total})$, or $F_{lat}(\theta_{total})$; 0.69 ± 0.02 vs 0.61 ± 0.03, 0.56 ± 0.03, and 0.63 ± 0.02, respectively, mean fraction correct ± SEM across 6 sessions; one-tailed paired *t* tests, all $p < 0.01$]. During contact between whisker and pole, M_0 and F_{ax} increase at different rates with increasing protraction. M_0 increases approximately linearly with protraction (Fig. 8*B*). In contrast, F_{ax} increases nonlinearly; F_{ax} becomes appreciable only after the whiskers are bent by interactions with the pole (Fig. 8*C*, Eq. 6). Because intrinsic whisker curvature in the plane of protraction was small ($\kappa_p = -5.6 \times 10^{-4}$; compare peak values during contact, $\kappa_p = 4.2 \times 10^{-3}$), F_{ax} builds up with a delay compared with M_0. For a conical whisker (Fig. 5), the bending stiffness (EI_p) is a steep function of distance from the face (Eq. 3). Forces applied by more distal objects will therefore produce more pronounced whisker bending and thus larger F_{ax} relative to M_0 for a given degree of protraction. As a consequence, each object location corresponds to a unique curve in the F_{ax} relative to M_0 plane, determined entirely by whisker mechanics and independent of object properties (Fig. 10*C*,*D*). Mice could therefore measure F_{ax} as a function of M_0 (Fig. 10*C*,*D*) to extract object location.

Discussion

Object localization is a fundamental haptic behavior. In addition to localization in the anterior–posterior direction (Knutsen et al., 2006; Mehta et al., 2007; O'Connor et al., 2010a; O'Connor et al., 2010b), rodents could use the amplitudes of time-varying stresses in the follicle to extract radial object location along the shaft of the whisker (Solomon and Hartmann, 2006; Szwed et al., 2006; Birdwell et al., 2007). In an aperture discrimination task, progressive whisker trimming dramatically reduced performance, suggesting that contact with multiple whiskers may be required for radial object localization (Krupa et al., 2001; Fig. 1*C*). However, the motor strategies underlying aperture discrimination remain unknown. The same whiskers may have contacted the aperture edge multiple times during head movements or follicle movement, which would provide azimuthal cues about the location of the aperture edge (Fig. 1*D*). Those experiments therefore did not reveal whether rodents can judge radial object distance based on forces measured in one follicle (Fig. 1*B*) or if they triangulate the distance to the aperture edge using multiple whiskers or by integrating over multiple contacts (Fig. 1*C*,*D*). Our experiments show that mice perform force-dependent measurement of radial distance and, more generally, can perform force measurement for tactile perception.

We found that mice could perform force-dependent radial object localization to at least 1 mm (Fig. 2*E*,*F*). Object localization typically involved multiple contacts between whisker and object. Individual contacts lasted 5–100 ms with peak forces up to 500 μN. The number and duration of contacts, the azimuths at contact, and the amplitudes of the forces as a function of protraction varied greatly across trials; however, these parameters were not correlated with behavioral performance (Figs. 6*F*–*I*, 7*B*, 9). Even single contacts were sufficient for discrimination in some trials. Precise timing of contact events within the whisk cycle does not seem to be critical for radial distance discrimination, because an ~5 ms uncertainty in the time of contact introduced by azimuthal jitter had no impact on the performance (Fig. 7*G*). Because the whisker contacted the pole at the same azimuthal angle, our experimental conditions exclude possible roles for protraction-dependent torsional cues (Knutsen et al., 2008).

Our calculations of the absolute magnitudes of the contact forces between whisker and object have to be viewed as estimates. The sizes of mouse whiskers change over days (Ibrahim and Wright, 1975), together with changes in the second moment of inertia and thus the bending stiffness (Eq. 3). However, we used the mechanical properties of representative whiskers to extract forces from curvature measurements for all whiskers. The mechanical model of the whisker was imperfect (see Materials and Methods; Birdwell et al., 2007; Quist et al., 2011). Estimates of the location of force measurement (p in Eqs. 2 and 3) were also approximate for two reasons. First, the C2 whisker has intrinsic curvature that is not localized to the plane of whisking (Quist and Hartmann, 2012 and our unpublished observations). Second, the fur on the face obscured the exact location of the end of the whisker. The precision of the force measurements is likely no better than a factor of 2–3. More accurate force estimates will require three-dimensional whisker tracking and mechanical models (Quist and Hartmann, 2012).

We varied the compliance of the object to confound the relationship between stresses in the follicle and object location (Fig. 9). Remarkably, increasing object compliance did not confuse the mice, even though stresses on the whisker follicle were greatly reduced. This suggests that the value of individual stresses as a function of protraction, or their time derivatives, are not solitary cues underlying coding of radial object distance. Instead, our data argue that mice effectively compare multiple variables at the follicle.

Although our behavioral data (Figs. 6, 9) allow for the possibility that mice rely solely on lateral force, we consider this possibility unlikely (Figs. 9, 10). The total lateral stresses are likely dominated by the bending moment due to mechanical advantage (Eqs. 1, 7). Assuming distances between the fulcrum within the follicle and any given mechanoreceptor in the range of 20 to 400

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(*Figure legend continued.*) consistent with the mice perceiving distal objects as proximal objects. Alternatively, the mice might have been generally confused, confounding the interpretation of the whisker stiffening experiment. Several lines of evidence argue against this. Stiffening whiskers on the contralateral side did not impair the mice in performing the task (unpaired *t* test, $p < 0.05$), ruling out nonspecific effects of the whisker manipulation. Additional evidence arguing that the mice continued to perform the object location task with stiffened whiskers is presented in the next panels. ***H***, ***I***, We tested other metrics of mouse behavior for evidence of confusion. Reaction times in hit trials for two mice with stiffened whiskers (gray) and with the stiffened C2 whisker on the contralateral side (black). We scored touches in 100 trials of one session of one mouse with a stiffened whisker (mean reaction time, 0.4 s). In 79% of trials, the mouse touched the pole at least once before making its decision. If mice stopped trying to solve the task and rather were impulsively licking to get water rewards, we would expect the mice to start licking as soon as the pole drop was audible (0 s). ***J***, Mean reaction time in hit trials in sessions with stiffened whiskers (gray) and sessions without whisker stiffening or stiffening of the contralateral C2 whisker (black).

μm, bending moments on the order of 1 μNm (Fig. 8*B*) will produce forces between 2.5 and 50 mN, 10- to 150-fold higher than peak lateral forces (Fig. 8*D*). Therefore, for mice to use lateral force as a sole cue, they would have to tease out lateral force from the much larger lateral stresses induced by bending moment (Fig. 10*B*). A better understanding of how forces are measured by the diverse sensory afferents in the follicle would help to disambiguate these mechanisms.

Our data further rule out that protraction-dependent bending moment by itself is used to determine radial distance (Birdwell et al., 2007). In contrast, a strategy in which axial stresses were compared with lateral stresses for a given protraction would be especially favorable. The whisker taper plays a critical role in this and related schemes. Bending stiffness is a steep function of whisker diameter. Because the taper of the whisker is linear (diameter from 70 μm at the base to 3–4 μm at the tip for a mature C2 whisker), the bending stiffness varies by a factor of almost 100,000 along the whisker (length, 16 mm; Eq. 3). Close to the whisker tip small forces are sufficient to bend the whisker, whereas large forces are required close to the follicle. As the whisker bends, axial forces, pushing the whisker into the follicle, build up. For more distal object locations, axial forces are relatively larger for a given bending moment. Mice could therefore extract object location by comparing lateral and axial stresses independently of their absolute values (Fig. 10*C*,*D*). Consistent with this view, stiffening the whisker and eliminating its taper appeared to confuse mice and make them interpret distal objects as proximal objects (Fig. 10*E*–*J*). Mice therefore appear to use the gradual whisker taper (Ibrahim and Wright, 1975; Fig. 5*A*, *B*) as a ruler to estimate radial object location.

Coding radial distance by relying on the relative amplitude of multiple force variables provides a robust algorithm for object localization. Mice are faced with measuring object location for objects with different properties, including different stiffness (i.e., another mouse compared with a rock). In addition, they may have to determine object distance under conditions in which contact forces between whisker and object cannot be controlled accurately (i.e., while running through a tunnel). Algorithms relying on relative amplitudes (e.g., Fig. 10*C*,*D*) can deduce object location independently of the compliance of the contacted object and whether it is moving.

There is one additional advantage of algorithms based on relative force measurements. Absolute forces can only be interpreted in terms of object location if the mouse has detailed knowledge of whisker protraction or has precise control of whisking. In contrast, coding of object distance by comparing two or more forces requires neither knowledge of whisker position nor precise control of whisking.

It has long been known that lateral stresses applied to whiskers excite trigeminal ganglion (TG) neurons in a direction- and velocity-dependent manner (Zucker and Welker, 1969; Gibson and Welker, 1983; Szwed et al., 2003; Szwed et al., 2006; Stüttgen et al., 2008). However, TG neurons also readily encode axial forces (Stüttgen et al., 2008). It is possible that different TG neurons are tuned to selectively detect lateral or axial stresses acting on the follicle and could thus underlie radial object localization.

Radial object localization with a single whisker presents a challenge for mice. Compared with object localization in the anterior–posterior direction, reaction times were longer (640 vs 470 ms on average) and the average number of contacts per trial was larger (7.0 vs 4.8; correct Go trials; A.H., D.H.O., and K.S., unpublished data). Learning was also relatively slow: at least 14 daily behavioral sessions were needed to train mice to perform the 3 mm radial distance discrimination task with a single whisker (C2) and an accuracy of >90% for at least 100 consecutive trials, longer than training for anterior–posterior object localization (compare Fig. 3 in O'Connor et al., 2010a). Further, in an aperture discrimination task, progressive whisker trimming dramatically reduced performance (Krupa et al., 2001). These observations indicate that, under natural conditions, rodents likely also rely on multi-whisker cues such as triangulation to judge object distance in three dimensions. Similarly, classic human psychophysics has shown that accuracy and speed in haptic object recognition increases with the number of fingers involved (Davidson, 1972).

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